Miyazaki Industrial AI Copilot SRS: Section-by-Section Explainer

Oct 24, 2026 · @Varizz

How to use this guide

This guide walks through the whole SRS ("Miyazaki Industrial AI Copilot — Software Requirements Specification", Draft v0.1, Oct 23, 2026, by @ATL Dev / Atomic Loops) in the same order as the SRS, one section at a time. Nothing is skipped: every section, epic, user story, tool, entity, requirement and appendix is covered.

The Word file does not expose page breaks, so "page by page" is done as "section by section". Each section of the SRS is roughly one to two printed pages, so you can still match this guide to the document you are holding.

Rules this guide follows

- Explanations are in plain words. Anything that is a real specification stays exactly as written in the SRS and is shown in `code style`: routes such as `/w/{company}/chats/{id}`, IDs such as `US-C04` and `NFR-08`, tool names such as `fetch_site(url, limits)`, numbers such as 12 characters, 7 days, 5 minutes.
- Each section ends with a **What matters most** note: the points a developer building the proof of concept (POC) must not get wrong.
- Where the SRS itself is unclear or leaves a decision open, this guide says so instead of guessing. The SRS lists its own open decisions in section 12.

Priority labels you will see (MoSCoW)

Label	Meaning	Release
Must	Required	v1
Should	Important, expected	v1
Could	Nice to have	v1.1
Won't (v1)	Deliberately postponed	later (v2)

One-sentence summary of the product (to keep in mind while reading): a Japanese-first, fully bilingual (Japanese and English) AI copilot where each manufacturing company gets its own isolated workspace; the AI reads the company's website and products, builds a verified company profile, asks follow-up questions through workflows, chats, drafts documents from a knowledge base, and helps the company find and plan AI use cases for its factory. A government-backed organisation (the Operator) manages all the companies from an admin console.

SRS Section 1: Introduction

What this section does: it sets the frame. It says what the system is, who the document is for, why the project exists, what is inside and outside the first release, and the vocabulary used everywhere else.

1 (opening paragraph): what is being specified

The system is a **multi-tenant, Japanese-first generative-AI copilot**. "Multi-tenant" means one platform serving many companies, each one walled off from the others. It would be run by the Miyazaki Prefecture industrial support organisation (called the **Operator**) for the manufacturing companies it supports. The document is the agreed baseline between Atomic Loops (the builder) and the Operator, written down before design and build so that scope, roles and behaviour are fixed in writing rather than discovered in code.

1.1 Purpose

Three kinds of readers use this document:

- **The Operator** confirms what the system will and will not do.
- **Atomic Loops' product, design and engineering team** designs, estimates and builds from it.
- **Whoever accepts the delivered system** uses the user stories and non-functional requirements as the acceptance checklist.

It covers: user roles, the page map, functional requirements (user stories with acceptance criteria), page examples, the AI agent and workflow specification, the data model, non-functional requirements and phased release scope. It does **not** cover commercial terms, hosting contracts or detailed visual design.

1.2 Background (why this exists, and the nine-part brief)

Miyazaki Prefecture runs a government-funded industrial area. Manufacturing companies go there for support, and the support organisation gives member companies benefits such as new connections and contacts. During Atomic Loops' visit in October 2026, one opportunity came out of the discussion: an AI copilot that the organisation operates for its member companies.

The brief has **eight parts from the discussion plus a ninth added on 23 October 2026** (Japanese and English throughout). Appendix E maps each part to the user stories that deliver it. In plain words, the nine parts are:

#	Brief item	What it means
1	Super-admin and logins	The Operator has super-admin access and creates a login (workspace) for each new company that joins.
2	Website and product analysis	A company enters its website and product details; the AI analyses the products and their potential.
3	Follow-up questions and workflows	The AI asks follow-up questions and creates follow-up workflows so the picture of the company gets more detailed over time.
4	AI-generated company page	The company's own page is generated by AI and holds the information about the company.
5	Many chats, workflow view, voice	One workspace holds many chats; a user can create any chat, see the workflow behind it, send messages and talk to the AI by voice.
6	Knowledge base and documents	The company gives prompts and the AI creates documents, so the system works as a copilot for the business and business planning.
7	Feedback everywhere	An option to give feedback on every page and output.

#	Brief item	What it means
8	Manufacturing AI use cases	A section where the company learns which AI use cases are possible in its plant, built on Atomic Loops' audit workflow (AI at every stage of the factory, plus upstream product discovery and product-market fit); the company defines and tracks its own use cases.
9	Japanese and English throughout	Added 23 Oct 2026. Japanese-language interface and outcomes everywhere, English as an equal second language: every dashboard element, chat and generative output works in both.

1.3 Scope

In scope for v1:

- A web application for desktop and mobile browsers, with an Operator admin console and **one isolated workspace per member company**.
- The AI agent services behind company analysis, follow-up questions, workflows, chats, document generation and use-case recommendations.
- A fully bilingual product (Japanese and English): every interface element, dashboard, email, AI reply and generated output exists in both languages (this is Epic K).
- The manufacturing AI use-case library and the explainer content on the Atomic Loops method, in both languages.
- Feedback capture on every AI output and every page, with an Operator-side inbox.

Out of scope for v1 (candidates for later, see section 11):

- Reading data from member companies' ERP, MES or machines, and deploying or running AI use cases. The platform **plans** use cases; it does not **execute** them in the plant.
- Native mobile apps. The web app is responsive instead.
- Matching companies to prefecture programmes, grants and partner introductions (v2).
- Public, unauthenticated pages and self-service sign-up. **Access is by invitation only.**
- Payments, pricing or commercial modules.

1.4 Definitions (the vocabulary of the whole SRS)

Learn these terms; every later section uses them.

Term	Plain-words meaning
Operator	The Miyazaki Prefecture industrial support organisation that runs the platform and onboards companies.
Member company	A manufacturing company supported by the Operator that gets a workspace.
Workspace	The isolated tenant for one company: its profile, chats, workflows, documents and use cases.
Super Admin	Operator staff member who can create workspaces, invite users and see the portfolio.
Platform Admin	Atomic Loops administrator who maintains the use-case library, workflow templates, prompts and models.
Company Admin	The first user of a workspace (typically the president or a director); manages members and sharing.
Member	Any other user inside a company workspace.
Coordinator	An Operator advisor to whom the company may grant read or comment access to its workspace.
Company profile	The AI-generated, user-verified page holding everything the platform knows about the company.
Fact	One statement in the profile, together with where it came from (provenance), how sure the AI is (confidence) and its review status. Provenance is one of: stated by a user, sourced from a document or website, or inferred by the AI.
Discovery workflow	A multi-step sequence the AI creates for a company (questions, uploads, analysis, review, output) to go deeper on one topic.

Term	Plain-words meaning
Step	One unit of a workflow, with an owner, a status and an output.
Chat	A conversation thread with the AI inside a workspace, optionally linked to a workflow or document.
Knowledge base	The company's sources (uploaded files, crawled pages, saved chats) and the documents generated from them.
Document	A file the AI drafts from a prompt plus the knowledge base; the company edits and exports it.
Use case	One AI opportunity in the company's plant or business, taken from the library or created by the company.
Horizon	H1 quick wins, H2 strategic bets, H3 disruptive solutions: the sequence of the Atomic Loops road map.
Autonomy level	L0 manual up to L5 full autonomy: a ladder describing who decides and when.
Entry point	How a use case gets its data: software only, camera based, or machine IoT.
Atomic Assessment	Atomic Loops' five-stage factory audit: KPI alignment, stakeholder interviews and Gemba, machine workflow, document review, processing and report.
Lite assessment	The self-service, AI-guided version of that audit inside the platform (v1.1).

1.5 References

The documents this SRS builds on: the Atomic Loops deck "Atomic Assessment: from quick wins to autonomous operations" (five stages, three outcomes, four steps, L0 to L5); the Atomic Loops "Factory Assessment Framework" (seven operational stages with human operations, software systems and AI opportunities at each, plus the engagement

framework); Japan's **Act on the Protection of Personal Information (APPI)**; **WCAG 2.1 level AA** and **JIS X 8341-3:2016** (web accessibility); **OWASP ASVS level 2** (application security).

1.6 Conventions

- MUST, SHOULD and MAY carry their usual meaning. "The AI" means the platform's own agent, not a third-party assistant.
- Numbering: user stories by epic (`US-A01` and so on), non-functional requirements `NFR-01` and so on, diagrams `D1` and so on.
- Priority uses MoSCoW. Must and Should are in v1, Could is v1.1, Won't (v1) is a later release. The Priority column in section 4 is a dropdown so the team can re-prioritise inside the document.
- Three provenance labels are used everywhere: **Stated, Sourced, Inferred**. Nothing the AI inferred is ever shown as a fact without that label.

What matters most

- Access is invitation-only; there is no sign-up page.
- The platform **plans** AI use cases but never connects to a factory's ERP, MES or machines.
- Stated / Sourced / Inferred is the core trust mechanism of the whole product.
- Japanese and English are both required from day one, not a later add-on.

SRS Section 2: Overall description

What this section does: it describes the product at a high level (vision, who uses it, where it runs, what is assumed, what limits it) before the detailed requirements begin.

Opening paragraph

The product turns the Operator's **first conversation with a company** into a **living, AI-maintained understanding of that company**, then uses it to plan the company's next steps: products, markets, the business plan and AI in the plant. The Operator sees its whole portfolio in one place. Atomic Loops' audit method is taught and applied through the use-case explorer.

2.1 Vision and value by stakeholder

Stakeholder	What they get
Member companies	A copilot that already knows their business, answers in Japanese at any hour, drafts documents they would otherwise write by hand, and shows concretely where AI fits in their plant.
The Operator	A consistent onboarding path for every company, a portfolio view of who needs what, and a stronger basis for the introductions and connections it makes.
Atomic Loops	Its method applied at scale; companies that arrive at an Atomic Assessment with a verified profile and a defined use-case list; feedback that improves the library.

2.2 Product context

In v1 the AI **never acts on the Operator's or a company's systems directly**. It reads only what users give it (websites, documents, answers) and writes only **inside the workspace**. Any action with a side effect, for example creating a workspace through the Admin Copilot, always needs a **human confirmation**.

2.3 Personas

Persona	Who they are	What they want	What slows them today	Device and language
Super Admin	Programme coordinator at the Operator	Create a workspace in minutes; see every company's status; route feedback and requests	Onboarding is a series of meetings and emails; no single view of member companies	Desktop; Japanese, some English

Persona	Who they are	What they want	What slows them today	Device and language
Company Admin	President or director of a small or mid-sized manufacturer	A clear picture of the company's position and next steps; documents for banks, customers and the prefecture; a concrete view of where AI applies	Little time, limited IT staff, planning in spreadsheets and on paper	Desktop and phone; Japanese only
Member	Production, quality, planning or sales manager	Answer follow-up questions about their area; draft specific documents; define plant use cases	Knowledge sits in people and on paper; no time to write it up	Desktop, and phone on the floor; Japanese
Coordinator	Operator advisor assigned to several companies	Read a company's profile and workflows before a meeting; comment; suggest next steps	Relies on notes from past visits	Desktop; Japanese
Platform Admin	Atomic Loops product and AI team	Maintain the library, templates, prompts and evaluation sets; watch quality and cost	Content changes need a deploy unless editable in the console	Desktop; English

2.4 Operating environment

A hosted web application in a **cloud region in Japan** (data residency is open question `OQ-03`). Supported browsers: the current and previous major versions of Chrome, Edge, Safari and Firefox on desktop; Safari on iOS and Chrome on Android on phones. **Nothing is installed** on member companies' computers.

2.5 Assumptions and dependencies

- The Operator creates every account; **no public sign-up**. One Company Admin per company is invited first, then invites colleagues.
- Member company websites are public and crawlable. Product details may also arrive as text, PDF catalogues, spreadsheets or photographs.
- Most member companies are small or mid-sized with limited IT staff, so the interface must be simple, Japanese by default and usable on a phone.
- A large-language-model provider is available with strong Japanese capability, a contract that **excludes training on customer data**, and a processing region acceptable to the Operator (`OQ-04`).
- Atomic Loops supplies and maintains the use-case library, workflow templates and question banks; the Operator reviews the Japanese wording of that content.
- Speech-to-text and text-to-speech with Japanese support are available for the voice feature.

2.6 Constraints (hard rules for the whole system)

- **Tenant isolation is absolute:** no user, model call or retrieval ever crosses from one company's workspace into another's.
- The AI performs **no action with a side effect** (creating, sending, deleting) without an **explicit confirmation** in the interface.
- All AI output carries **provenance**; numbers that were neither stated nor sourced are labelled as **estimates**.
- Compliance with **APPI** for any personal data (user accounts, names inside documents), plus an **audit log** of administrative actions.
- The interface and all AI responses work fully in Japanese; English is a second, equal language.

2.7 Release phasing

Section 11 defines three releases: **v1** (the minimum the Operator can onboard companies with), **v1.1** (voice, planning canvases, the lite assessment) and **v2** (programmes and connections, integrations). The priorities in section 4 follow this split.

What matters most

- Two non-negotiables: tenant isolation and confirmation before any side-effect action.
- The AI can only read what users give it and write only inside the workspace.
- The primary users are non-technical Japanese manufacturing executives, often on a phone: keep the interface simple.

SRS Section 3: Information architecture

What this section does: it defines every page (with its URL route), how users move around, who may open what, and the main user journeys. This is the blueprint for your routing and permissions.

Opening paragraph

The app has **two areas**: the Operator's **admin console** and **one workspace per member company**, both inside one shared shell (search, notifications, language, feedback). A Super Admin or Coordinator can switch between company workspaces from the top bar. A company user only ever sees their own workspace.

3.1 Page map (all routes, exactly as specified)

Area	Route	Page	Purpose	Who sees it
Public	`/login`, `/invite/{token}`, `/reset`	Sign-in, invitation, password reset	Entry to the platform; the invitation page sets language and password	Everyone
Admin	`/admin`	Portfolio dashboard	Companies by status, onboarding funnel, activity, feedback sentiment	Super Admin, Platform Admin
Admin	`/admin/companies`	Companies list	Status, profile completeness, last activity; create, invite, deactivate	Super Admin

Area	Route	Page	Purpose	Who sees it
Admin	`/admin/companies/{id}`	Company record	Shared profile summary, users, workflow and use-case counts, requests, sharing settings	Super Admin, assigned Coordinator
Admin	`/admin/users`	Users	All Operator and company users, roles, MFA status	Super Admin
Admin	`/admin/library`	Use-case library	Create, edit, translate and publish use cases, stages and explainer content	Platform Admin; Super Admin reviews
Admin	`/admin/templates`	Workflow and document templates	Question banks, step definitions, document templates	Platform Admin
Admin	`/admin/feedback`	Feedback inbox	Triage, tag, reply, export	Super Admin, Platform Admin
Admin	`/admin/copilot`	Admin Copilot	Natural-language admin actions with confirmation (v1.1)	Super Admin
Admin	`/admin/audit`	Audit log	Who did what, when	Super Admin, Platform Admin
Admin	`/admin/settings`	Settings	Organisation name, branding, default language, retention	Super Admin
Workspace	`/w/{company}/home`	Home	Profile completeness, open steps, recent chats, suggested next actions	All company roles
Workspace	`/w/{company}/profile`	Company profile	The generated, verified profile by section; products; change history	All company roles; Coordinator sees shared sections
Workspace	`/w/{company}/workflows`	Workflows	Active, proposed and completed workflows; create from template or from a goal	All company roles
Workspace	`/w/{company}/workflows/{id}`	Workflow	Stepper, step detail, linked chat, dependency view	All company roles
Workspace	`/w/{company}/chats`	Chats	List, search, pin; new chat	All company roles
Workspace	`/w/{company}/chats/{id}`	Chat	Messages, citations, context panel, voice	All company roles
Workspace	`/w/{company}/knowledge`	Knowledge base	Sources tab, Documents tab, Generate tab	All company roles

Area	Route	Page	Purpose	Who sees it
Workspace	`/w/{company}/knowledge/documents/{id}`	Document	Editor with AI actions, versions, export	All company roles
Workspace	`/w/{company}/planning`	Business planning (v1.1)	Canvases: business model, product-market fit, go-to-market, financial outline	Company Admin, Member
Workspace	`/w/{company}/ai-use-cases`	AI use-case explorer	Method, factory map, library, my use cases, readiness, road map, request	All company roles
Workspace	`/w/{company}/ai-use-cases/{id}`	Use case	Charter, readiness check, prerequisites, feedback	All company roles
Workspace	`/w/{company}/feedback`	My feedback	Everything the user submitted and its status	All company roles
Workspace	`/w/{company}/settings`	Workspace settings	Name, members, language, sharing with the Operator, data export and deletion	Company Admin
Global	(shell)	Search, notifications, help, feedback widget	Available on every page	Everyone

3.2 Navigation model

- **Left rail:** workspace sections (Home, Profile, Workflows, Chats, Knowledge, Planning, AI use cases). The admin console has its own rail.
- **Top bar:** company switcher (admins and coordinators only), global search, notifications, language toggle, user menu.
- **Right-hand context panel:** opens on chat, workflow and document pages. It shows what the AI is using (profile, sources, workflow step) and the feedback controls.
- **"Ask the copilot" button:** persistent on every workspace page; it starts a chat with the current page as context.
- **Feedback widget:** bottom-right corner of every page.

3.3 Role and page access

Page group	Platform Admin	Super Admin	Coordinator	Company Admin	Member
Portfolio dashboard, companies, users	Full	Full	Assigned companies, read	None	None

Page group	Platform Admin	Super Admin	Coordinator	Company Admin	Member
Library, templates, prompts	Full	Review and comment	None	None	None
Feedback inbox, audit log	Full	Full	None	None	None
Company profile	Read, for support only	Shared sections only	Shared sections, comment	Full	Full
Workflows, chats, knowledge base, planning	None	None	Read and comment, if granted	Full	Full
AI use-case explorer	Content only	Content only	Read, if granted	Full	Full
Workspace settings	None	Deactivate only	None	Full	Own profile only

"**Shared sections**" are the profile sections the Company Admin has chosen to share with the Operator. The default is the **overview** and the **products summary**.

Notice what this table says: the Operator (Super Admin) can **not** see a company's chats, workflows, knowledge base or planning at all. The Coordinator can read and comment on them only if the company grants it.

3.4 End-to-end workflow and key user journeys

A diagram "D3" in the original shows the whole workflow end to end, numbered in the order a company meets it (the diagram is not in the text version of the file; the journeys below are its written paths). All journeys follow one pattern: **AI proposal, then human review, then a recorded outcome**.

- **J2, deepening the profile:** a Member answers a workflow's questions by text or voice; the AI proposes profile updates as Stated facts; the Company Admin confirms them; the completeness score rises.
- **J3, creating a document:** a user opens the Generate tab, picks a template or writes a prompt, approves the outline, edits the draft with inline AI actions and exports it. The document can be saved back as a source.
- **J4, finding AI use cases:** a user reads the method explainer, opens the factory map, accepts AI-matched recommendations into "My use cases", runs the readiness check, places them on the road map and, if they wish, requests an Atomic Assessment.

- **J5, operating the portfolio:** a Super Admin reviews the dashboard, opens a company record, reads and answers the feedback inbox, and uses the Admin Copilot to create the next workspace.

Note: the SRS text lists J2 to J5 only. A first journey (J1, presumably onboarding a company) is not written out in this section; it is probably in the D3 diagram. This is worth asking the document owner about.

What matters most

- Routes are fixed: admin under `/admin/...`, company workspaces under `/w/{company}/...`.
- Permissions are enforced per role per page group (table 3.3), and enforced on the server (see `NFR-07`).
- The Operator sees only what the company chooses to share; default shared sections are overview and products summary.
- Every journey follows: AI proposes, human reviews, outcome is recorded.

SRS Section 4: Functional requirements

How section 4 works: each requirement is a **user story** ("As a [role], I want [thing], so that [benefit]") with **acceptance criteria** (the checkable conditions that mean the story is done) and a **priority** (Must, Should, Could, Won't (v1)). The stories are grouped into **ten epics, A to J**, in the order a company meets them; an eleventh epic **K** covers Japanese and English. Section 5 shows the pages the stories land on, and Appendix E traces them back to the nine parts of the brief.

In the tables below, the "Exact criteria" column keeps the SRS's numbers and rules; the "Plain meaning" column is explanation only.

4.1 Epic A: access, tenancy and roles

Epic purpose: how workspaces and users get created, how people sign in safely, and how one company's data is kept away from another's.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-A01`	Must	The Operator's Super Admin creates a company's workspace by entering its name, industry, size band, contact person and default language. Onboarding takes minutes.	Workspace created with a **unique slug**; appears in the companies list with status **Invited**; the action is in the **audit log**.
`US-A02`	Must	The Super Admin invites the company's first user (the Company Admin) by email so the company can start on its own.	Invitation email in the workspace language, link valid **7 days**; **resend and revoke** available; status moves to **Onboarding** when accepted.
`US-A03`	Must	The invited person accepts, sets a password and picks a language to enter the workspace.	Password policy **12 characters or more**; language saved to the user; the **onboarding wizard opens on first entry**.
`US-A04`	Must	The Company Admin invites colleagues as Members so that the people who know each area can contribute.	Invite by email with role Member; remove a member; pending invitations listed with resend.
`US-A05`	Must	Basic account safety for everyone: sign-in, sign-out, password reset, idle timeout.	Reset link valid **1 hour**; session ends after **12 hours idle**; repeated failed sign-ins are **rate-limited and logged**.
`US-A06`	Should	Multi-factor authentication (MFA) with an authenticator app, required for admin roles.	Enrolment **forced at first sign-in** for Super Admin and Platform Admin; **recovery codes** issued; MFA status shown on the users page.
`US-A07`	Must	The platform itself must keep every workspace's data isolated so no company sees another's.	Every query **scoped by workspace on the server**; automated tests prove cross-workspace requests are refused; **retrieval indexes partitioned per workspace**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-A08`	Should	A Company Admin can grant a named Coordinator (Operator advisor) read or comment access.	Grant and revoke from workspace settings; the grant is in the audit log; the Coordinator can **read and comment but not edit**.
`US-A09`	Must	Any user can switch the interface between Japanese and English.	Interface, emails and AI replies follow the user's language; a user can ask the AI for **one reply in the other language**.
`US-A10`	Must	The Super Admin can deactivate a workspace (a company leaves the programme) and reactivate it later without losing data.	Deactivated users **cannot sign in**; data kept for the **retention period**; reactivation restores everything unchanged.

Company status lifecycle seen here and in Epic B: Invited (workspace created) then Onboarding (invitation accepted) then Active, with Dormant (no activity for 30 days) and Deactivated as side states.

What matters most

- There is a fixed chain of account creation: Super Admin creates workspace, invites Company Admin, who invites Members. No self-service sign-up.
- Security numbers to remember: 12-character passwords, 7-day invites, 1-hour reset links, 12-hour idle timeout.
- `US-A07` (tenant isolation) is the most important story in the epic; it has automated tests as part of its acceptance criteria.
- Two similar-looking 12s: password length 12 characters, idle session 12 hours. They are different rules.

4.2 Epic B: admin console

Epic purpose: the tools the Operator (Super Admin) and Atomic Loops (Platform Admin) use to run the platform: company lists, the portfolio dashboard, the use-case library, templates, the feedback inbox, the audit log, the Admin Copilot and announcements.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-B01`	Must	A list of all companies so the Super Admin can see who needs attention: status, how complete the profile is, how many use cases, when last active.	Statuses **Invited, Onboarding, Active, Dormant (no activity for 30 days), Deactivated**; **filter, sort and CSV export**.
`US-B02`	Must	A company record page to prepare for a conversation with that company.	Shows **only sections the company has shared**; counts never expose content; links to the feedback and requests from that company.
`US-B03`	Must	A portfolio dashboard for reporting on the whole programme.	Counts by status; **onboarding funnel** (invited, accepted, profile approved, first workflow done); **weekly active companies**; industries; most-chosen use cases; feedback sentiment; all **aggregated and respecting sharing settings**.
`US-B04`	Must	Platform Admin manages the use-case library so content improves without a software deploy.	Create and edit use cases with **Japanese and English fields**, stage, entry point, horizon, autonomy level and data needed; states **draft, review, published**; **version history**.
`US-B05`	Should	Platform Admin manages workflow templates and question banks.	Steps, questions, branching rules and outputs editable in both languages; **publishing creates a new template version**; **running workflows keep their version**.
`US-B06`	Must	A feedback inbox so every piece of feedback is seen and answered.	All feedback with type, page and a reference to the AI output; statuses **New, In review, Addressed**; tags; reply to the user; export.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-B07`	Must	An audit log for accountability of admin actions and of any Operator access to company data.	**Immutable** entries with actor, action, target and time; filter by actor, company and date; **retained for 1 year**.
`US-B08`	Could	An Admin Copilot that carries out admin requests written in plain language.	AI shows the **exact action** it will take; **nothing executes before confirmation**; every action logged; portfolio questions answered from **aggregated data only**.
`US-B09`	Could	Announcements to every company (events, deadlines).	Shown on every workspace home; included in the email digest; can be **scheduled and withdrawn**.
`US-B10`	Won't (v1)	Record programmes, events and partners, and let the AI match companies to them (later).	**Deferred to v2**; companies **opt in** to matching; matches are suggestions the Operator confirms.

Where each story lives (from section 3): companies list at `/admin/companies`, company record at `/admin/companies/{id}`, dashboard at `/admin`, library at `/admin/library`, templates at `/admin/templates`, feedback inbox at `/admin/feedback`, audit log at `/admin/audit`, Admin Copilot at `/admin/copilot`.

What matters most

- Privacy by design: the Super Admin sees only what the company shares (`US-B02`, `US-B03`), and the counts on a company record never expose content.
- The library and templates must be editable **without a deploy** (`US-B04`, `US-B05`), with draft, review and published states and version history.
- Library content is bilingual by construction: both Japanese and English fields exist.
- The audit log is immutable and kept for 1 year; it also records Operator access to company data.
- Admin Copilot (`US-B08`) and announcements (`US-B09`) are Could, so v1.1. The template editor (`US-B05`) is Should but section 11 puts it in v1.1.

4.3 Epic C: onboarding and the company profile

Epic purpose: this is the heart of the product. The Company Admin enters a website and product details; the AI reads and analyses them and produces a **company profile** made of individual **facts**. Every fact shows where it came from and must be confirmed, edited or rejected by a human. This epic covers brief items 2 and 4.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-C01`	Must	An onboarding wizard collects the website, product details and basic facts so the AI can start the profile.	Fields for **website URLs, product text, catalogue and photo uploads, employees, revenue band, plants, certifications, industries served, customer types**; **progress saved; resumable**.
`US-C02`	Must	The AI reads the company website so people do not retype public information.	**robots.txt respected**; **depth and page limits** applied; every extracted fact carries its **source URL**; live progress; completes within **5 minutes for sites up to 200 pages**; a failed fetch shows a clear message and a **manual entry fallback**.
`US-C03`	Must	The AI analyses each product: function, process, customers, differentiators and potential, like a market analyst.	**One entry per product or product family** with provenance and confidence per statement; the user can **add, merge, rename or delete** products; adjacent applications and potential markets are labelled **Inferred**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-C04`	Must	The generated profile only keeps what is true: the Company Admin confirms, edits or rejects every fact.	Sections: **Overview, Products and services, Markets and customers, Manufacturing capabilities, Supply chain and partners, Competitive position, Potential and opportunities, Open questions**; each fact shows **Stated, Sourced or Inferred** and a **confidence**; **rejected facts leave the profile and the AI context**; every change logged with editor and time.
`US-C05`	Must	A completeness score and a gap list show what to answer next.	Score **per section and overall**; each gap names the **workflow or question** that would fill it; home page shows the **top three gaps**.
`US-C06`	Must	Facts that a Member states in chats, answers and documents flow into the profile after review.	**Proposed facts queue** with provenance; Company Admin confirms or rejects; **per-fact history with before and after values**.
`US-C07`	Should	The Company Admin can re-run the website analysis to follow site changes.	A **diff** of new, changed and removed facts is shown; **confirmed facts are never overwritten without review**.
`US-C08`	Must	The Company Admin chooses which profile sections the Operator can see.	**Default: overview and products summary**; changes logged; shared sections carry a **"visible to Operator"** marker.
`US-C09`	Should	Export the profile (for example to give to a bank or customer).	**DOCX and PDF**, in Japanese or English; the export carries the **as-of date** and the **provenance legend**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-C10`	Must	Security rule: website and document content is data, never instructions, so a crafted web page cannot change the AI's behaviour (protection against prompt injection).	Instruction-like text found in sources is **ignored and flagged on the source**; a **red-team test set of injected pages passes before each release**.

The three provenance labels (central idea)

Label	Meaning
Stated	A user said it (typed, spoken, answered).
Sourced	Taken from a document or website, with a link back to the source.
Inferred	The AI guessed or concluded it. Always shown with a label and a confidence; it becomes Stated only when a user confirms it (section 6.3).

What matters most

- The profile has exactly **eight sections** (listed in `US-C04`). Use the names as written.
- **Nothing the AI creates enters the profile as truth without human review.** Rejected facts are removed from the AI's own context too.
- Website crawling must respect robots.txt, have page and depth limits, record the source URL for each fact, and finish within 5 minutes for up to 200 pages.
- Prompt-injection defence (`US-C10`) is a Must, tested before every release.
- Facts are reached from several places: website analysis, workflow answers, chats, documents. They all go to the same review queue.

4.4 Epic D: discovery workflows

Epic purpose: a **workflow** is how the AI turns a gap in the profile into a short, structured piece of work for the company. The AI proposes it, the company runs it, and its outputs land in the profile, the knowledge base or the use-case list. This epic covers brief item 3 (follow-up questions and follow-up workflows).

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-D01`	Must	Once a profile exists, the AI proposes the first workflows so the company knows where to start.	Proposals: **Company discovery, one Product deep-dive per main product, Market and product-market fit, Manufacturing and AI readiness**; each with a **rationale** and the **number of questions and uploads** involved; the user **starts, defers or dismisses** each.
`US-D02`	Must	A workflow is a clear sequence of steps, each with a status and an owner.	Step types: **question set, upload request, AI analysis, review and approval, output document**; statuses **Not started, In progress, Waiting for review, Done, Skipped**; **owner** and optional **due date** per step.
`US-D03`	Must	The AI asks questions in small adaptive sets so answering feels like a conversation, not a form.	**5 to 10 questions per set**; follow-ups depend on earlier answers; answer by **text, voice or upload**; **"I don't know"** and **"assign to a colleague"** available; answers become **Stated facts in the review queue**.
`US-D04`	Must	A workflow can be seen as a stepper and as a dependency view.	**Stepper with progress**; **dependency view** shows which profile sections, documents and use cases each step feeds and which steps wait on which.
`US-D05`	Must	Every workflow has its own chat.	Messages carry the workflow context; the AI can **mark a step done or move to the next one from the chat after confirmation**.
`US-D06`	Should	A Company Admin can build their own workflow from a template or by describing a goal.	The AI **drafts the steps from the goal**; the user **reorders, adds, removes and assigns** steps before starting.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-D07`	Should	Reminders and pause/resume so workflows do not stall silently.	Notification when a step waits on me; **weekly digest** of open steps; **pause and resume keep all answers**.
`US-D08`	Must	A finished workflow delivers its outputs and proposes what comes next.	Outputs as defined by the template (**profile updates, a document, use-case candidates**); next workflows proposed with rationale.
`US-D09`	Should	Any chat can be turned into a workflow.	**"Turn this into a workflow"** drafts steps from the conversation; the user edits before starting.

Workflow vocabulary in one place

- **Step types (5):** question set, upload request, AI analysis, review and approval, output document.
- **Step statuses (5):** Not started, In progress, Waiting for review, Done, Skipped.
- **Four starter workflows proposed by the AI (`US-D01`):** Company discovery; Product deep-dive (one per main product); Market and product-market fit; Manufacturing and AI readiness. The full step lists of these and other templates are in Appendix B.

What matters most

- Workflows are **AI-proposed, human-run**. The user can start, defer or dismiss a proposal.
- Question sets are **5 to 10 questions**, adaptive (later questions depend on earlier answers), and every answer becomes a **Stated fact in the review queue**, never straight into the profile.
- Each workflow has exactly one linked chat; the AI may change step status from the chat only after the user confirms.
- Existing running workflows keep their template version even if the template is updated later (`US-B05`).
- v1 includes `US-D01` to `US-D05` and `US-D08` (section 11). `US-D06`, `US-D07`, `US-D09` are Should but fall into v1.1 in the release table.

4.5 Epic E: chats and voice

Epic purpose: many chats can live in one workspace. Each chat knows the company, can be linked to a workflow or document, shows what the AI is using, and can be spoken to by voice. This epic covers brief item 5.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-E01`	Must	A chat list that can be searched, pinned, renamed and archived; a chat can start blank, from a template, from a workflow or from a document.	List **sorted by last activity**; search across **titles and content**; a chat started from a workflow or document opens with that **context attached**.
`US-E02`	Must	Replies stream in with formatting and citations, and each message has useful actions.	Streaming starts within **2 seconds at p95**; markdown and tables render; citations link to profile facts, sources or web pages; actions: **copy, feedback, save to knowledge base, turn into document, turn into workflow, propose to profile**.
`US-E03`	Must	The user can see and control what context the AI uses, so they trust it.	**Context panel** lists profile sections, sources and workflow in use as **toggles**; the AI **states when it answers without the profile**.
`US-E04`	Could	Talk to the AI by voice (on the floor, or while driving).	**Push-to-talk and hands-free** modes; **Japanese and English** recognition; optional **spoken replies**; transcript saved as the message text; works in a **phone browser**.
`US-E05`	Must	Attach files in a chat so the AI can read a drawing, price list or report.	**PDF, images, spreadsheets, Word and PowerPoint**, up to **25 MB per file**; an attachment can be saved as a source.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-E06`	Should	Suggested prompts based on the company's situation.	Suggestions drawn from **open gaps, waiting steps and newly relevant library use cases**; dismissible.
`US-E07`	Must	Facts stated in chat are captured for the profile.	The AI **proposes recognised facts to the review queue with the message as source**; **nothing enters the profile without confirmation**.
`US-E08`	Should	Share a chat with colleagues and export it.	Share with members; export as **Markdown or PDF**; a shared chat shows **who said what**.
`US-E09`	Must	A monthly token budget per workspace so cost stays predictable.	Default set in admin settings; the workspace sees a **notice at 80%** and a clear message **at the limit**; the Super Admin can raise it.
`US-E10`	Must	Long chats keep working over weeks.	Older messages are **summarised into the context**; the summary is **visible in the context panel**.

Message actions on every AI reply (from `US-E02`)

Copy; Feedback (Helpful?); Save to knowledge base; Turn into document; Turn into workflow; Propose to profile. These appear as icon plus label, collapsing into a menu on phones (section 5.10).

What matters most

- Voice (`US-E04`) is **Could, v1.1**. Everything else in this epic that is Must is in v1. The chat UI must still be designed so a microphone can be added later (the wireframe already shows the mic button).
- A real latency target: first token within 2 seconds at p95.
- Context transparency is a feature: toggles in the context panel and a notice when the profile is not used.
- Token budget at workspace level: notice at 80%, hard message at the limit.
- Chat attachments: allowed types and 25 MB cap per file.

- Facts mentioned in chat do not auto-enter the profile; they go to the review queue.

4.6 Epic F: knowledge base and document studio

Epic purpose: the company adds its own files and web pages as **sources**; the AI then drafts **documents** from a prompt plus those sources; the user edits, exports and can save the finished document back as a source. This epic covers the first half of brief item 6.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-F01`	Must	Add files and web pages as sources so the AI knows what the company knows.	**PDF, DOCX, XLSX, PPTX, images, text and URLs**; **OCR for scanned Japanese PDFs**; statuses **Processing, Ready, Failed**; **automatic tags**; **1 GB per workspace in v1, configurable**.
`US-F02`	Must	Open a source and see what the AI extracted, to check and clean it.	Extracted text, **AI summary** and **detected facts** shown; **re-index** available; deleting a source **removes it from retrieval within one minute**.
`US-F03`	Must	Generate a document from a prompt and a template.	**Template or free prompt**; context chosen from **profile, selected sources and a workflow's answers**; **language and length** chosen; the request is **saved for re-use**.
`US-F04`	Should	Approve an outline before the full draft.	Outline shown first; sections can be **added or removed**; **each section can be regenerated on its own**.
`US-F05`	Must	An editor with AI actions on selected text.	Rich text; **rewrite, shorten, expand, translate between Japanese and English, insert profile facts, build a table from sources**; **comments**; **versions with compare and restore**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-F06`	Must	Export and share a document.	**DOCX, PDF and Markdown** export; share with members and, if granted, the Coordinator.
`US-F07`	Must	Save a finished document as a source so later chats and documents build on it.	One action adds the document to the knowledge base and **indexes it**; the source shows its **origin**.
`US-F08`	Should	Maintain document templates (Platform Admin) and save the company's own (Company Admin).	Platform templates in both languages; a **company template saved from any document**; templates **versioned**.
`US-F09`	Must	Every statement in a generated document shows whether it is sourced.	Sourced statements **link to the source**; unsourced statements carry an **"AI assumption, verify"** marker; the export can **include or exclude** the markers.
`US-F10`	Could	Translate a whole document between Japanese and English.	A **workspace glossary** keeps product and process terms consistent; the translation becomes a **new version**.

Where this appears

- Page `/w/{company}/knowledge` has three tabs: **Sources**, **Documents**, **Generate**.
- A single document opens at `/w/{company}/knowledge/documents/{id}` with the editor, AI actions panel, versions and export.
- The 12 document templates available in v1 are listed in Appendix D.

What matters most

- Source types, size cap (1 GB per workspace) and the three source statuses (Processing, Ready, Failed) are all specified.
- Deleting a source must remove it from retrieval within one minute. Section 6.3 adds: facts that came only from that source go back to review as Unsourced.
- The "AI assumption, verify" marker is the document-level version of the Stated/Sourced/Inferred idea.

- Generated documents target about 2,000 words in 90 seconds (`NFR-02`).
- In section 11, v1 includes `US-F01` to `F03` , `F05` to `F07` and `F09` ; `US-F04` , `F08` are Should; `US-F10` is v2.

4.7 Epic G: business planning copilot (v1.1)

Epic purpose: the second half of brief item 6, "a copilot for business planning". Planning **canvases** (structured boards) are pre-filled from the company profile, challenged by the AI, linked to evidence and turned into actions. Every story here is **Could**, so the whole epic is **v1.1**, not v1. The page is `/w/{company}/planning` .

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-G01`	Could	Planning canvases that are pre-filled from the profile, so planning starts from what the AI already knows.	**Business model canvas; product-market fit board (segment, problem, product, alternatives, evidence, status); go-to-market plan; three-year financial outline; export readiness**; each cell editable; AI fills carry provenance.
`US-G02`	Could	The AI challenges a canvas to find weak assumptions early.	A **"challenge" action** lists the **weakest assumptions, why, and the evidence that would resolve each**.
`US-G03`	Could	Each hypothesis links to its evidence, so the team can track what is proven.	Links to **sources, chats and workflow answers**; statuses **Untested, Validated, Invalidated**.
`US-G04`	Could	An action plan derived from the canvases.	**Tasks with owner and due date**; progress shown on home.
`US-G05`	Could	Export the whole planning pack.	**One document from all canvases** in the chosen language, through the document studio.

What matters most

- The whole epic is out of v1 (v1.1). You can leave the `/planning` route out of the first POC or show it as a placeholder.

- Still, v1 pieces already point to it: the Market and product-market fit workflow (Appendix B, B3) says its hypotheses go to the planning board in v1.1.
- Hypothesis status is a closed set of three: Untested, Validated, Invalidated.
- The product-market fit board has six named fields: segment, problem, product, alternatives, evidence, status.

4.8 Epic H: manufacturing AI use-case explorer

Epic purpose: this is where the **Atomic Loops method** lives inside the product (brief item 8). It teaches how AI fits a factory, shows use cases stage by stage, matches them to the company's profile and lets the company define its own, using the same horizon (H1 to H3) and autonomy (L0 to L5) language as an Atomic Assessment report. Use cases cover every stage of the factory flow and extend upstream into product discovery and product-market fit. The library is not a fixed list; it grows over time. Page: `/w/{company}/ai-use-cases` (and `/w/{company}/ai-use-cases/{id}` for one use case).

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-H01`	Must	A short explainer of how AI fits a factory, so the admin can talk about it with their team.	Covers the **five audit stages, the three outcomes of a report, the four steps (descriptive, predictive, prescriptive, autonomous), the L0 to L5 ladder, the three horizons and the value principles (measured before modelled, targets derived, haircuts applied, overlaps combined)**; each concept has **one example**; Japanese and English; **readable in under 15 minutes**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-H02`	Must	A clickable factory map, stage by stage.	**Seven operational stages (inward, warehouse, kitting, assembly, final inspection, packaging, logistics)** plus **lanes** for planning and supply chain, maintenance, quality, R&D and product discovery, and sales and product-market fit; selecting a stage shows **typical human operations, software systems and example use cases**; **stages indicated by our profile are highlighted**.
`US-H03`	Must	Each library use case is a searchable, filterable card.	Card fields: **title, stage, problem solved, how it works, data needed, systems touched, entry point (software only, camera based, machine IoT), horizon, autonomy level, prerequisites, what good looks like**; value shown **only as observed ranges with no client named**; Japanese and English; filters by **stage, entry point, horizon and level**.
`US-H04`	Must	The AI recommends use cases that fit the company's profile, so nobody has to read the whole library.	**Ranked list** with a **rationale tied to profile facts and the readiness gaps it would face**; **"add to my use cases" in one click**; **feedback on every recommendation**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-H05`	Must	The company defines its own use case with the AI's help, so ideas from the floor become plannable.	**Wizard**: problem, stage, who is affected, how data is captured today (paper, spreadsheet, system, sensor), systems involved, expected outcome and KPI; the AI drafts a **one-page charter**, **classifies the step and autonomy level** and **lists prerequisites and data needs**; the user edits and saves.
`US-H06`	Should	A readiness check per use case: what must exist before it can run.	**Four dimensions (data, process, people, systems)** with **3 to 5 questions each**; a readiness rating and gap list; the **level the use case needs against where we are today**.
`US-H07`	Must	A road map board to sequence use cases the way the method says.	A **2x2 of business impact against readiness to deploy**, quadrants **Quick wins, Strategic bets, Momentum builders, Later**; a **programme view by horizon (H1 to H3)** on a multi-year line; **drag and drop**; dependencies shown as **"needs the data created by"**; **export as a document**.
`US-H08`	Could	A self-service lite assessment to prepare before a full audit.	**Five stages** in self-service form: KPI alignment questions, a pain-point interview in chat, stage-by-stage capture of each line or machine with data / human / material lanes, a document checklist, and a **preliminary** readiness summary with recommended quick wins; **every output labelled preliminary**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-H09`	Could	Request an Atomic Assessment from inside the platform.	Form pre-filled from profile, use cases and readiness; **routed to Atomic Loops and the Operator**; statuses **Requested, Scheduled, In progress, Report delivered** visible to the company.
`US-H10`	Must	Learning content per stage, so users understand each use case before committing.	A **short explainer, an anonymised example and a "what we need from the plant" list per stage**; a **glossary in both languages**.

Method vocabulary used in this epic

Concept	Values
Horizon	**H1** quick wins, **H2** strategic bets, **H3** disruptive solutions
Autonomy level	**L0** to **L5** (names in Appendix F: Manual, Assisted, Partial, Conditional, High, Full)
Four steps of analytics	descriptive, predictive, prescriptive, autonomous
Entry point	software only, camera based, machine IoT
Atomic Assessment stages	KPI alignment, stakeholder interviews and Gemba, machine workflow, document review, processing and report
Road map quadrants	Quick wins, Strategic bets, Momentum builders, Later
Readiness dimensions	data, process, people, systems
Seven factory stages	inward, warehouse, kitting, assembly, final inspection, packaging, logistics
Lanes around the stages	planning and supply chain; maintenance; quality; R&D and product discovery; sales and product-market fit

What matters most

- The explorer has several views in one page: Method, Factory map, Library, Mine (My use cases), Road map (see section 5.7 tabs). Readiness and request are part of the page group in section 3.1.
- Value numbers are shown only as **observed ranges with no client named**.
- The AI never executes use cases in the plant; it classifies and plans them.
- v1 includes `US-H01` to `H05`, `H07`, `H10`. `US-H06` (readiness check), `H08` (lite assessment), `H09` (assessment request) are v1.1 per section 11.

4.9 Epic I: feedback everywhere

Epic purpose: brief item 7. Users can rate any AI output and send feedback from any page; the Operator and Atomic Loops see it in an inbox; feedback also improves the AI.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-I01`	Must	Rate any AI output in one click and say why.	**Thumbs up or down**, a **reason from a short list** and **free text** on every **message, profile fact, document, recommendation and charter**; **non-blocking**; stored with a reference to the output.
`US-I02`	Must	A feedback widget on every page to report anything.	Categories: **bug, idea, confusing, wrong, praise**; free text; **optional screenshot**; **page, role and language captured automatically**.
`US-I03`	Must	Report a wrong fact right where it appears.	The fact is marked **Disputed**; the correct value is requested; the item goes to the **review queue and the inbox**.
`US-I04`	Must	See what happened to your feedback, so giving it feels worthwhile.	Statuses **Received, In review, Addressed**; the reply from the Operator or Atomic Loops shown under the item.
`US-I05`	Must	Feedback analytics for the Super Admin.	Volume, **positive rate**, by feature and by company **in aggregate**, trend over time; export.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-I06`	Must	Real feedback and the related output feed the evaluation set so prompts and models can be tuned.	Available **only where the company's consent setting allows**; **never used to train third-party models**; **personal data removed before use**.
`US-I07`	Should	A short quarterly in-app survey.	**Three questions, once per quarter per user**, dismissible; results on the dashboard.

Note the two status vocabularies: the admin inbox uses **New, In review, Addressed** (`US-B06`), and the user sees **Received, In review, Addressed** (`US-I04`). The SRS does not state how they map; most likely "New" for the admin is the same stage as "Received" for the user. Confirm this with the document owner.

What matters most

- Feedback happens at two levels: **per output** (thumbs, reason) and **per page** (widget). Both end up in the same inbox.
- Reporting a wrong fact (`US-I03`) is a link back into the profile: the fact turns **Disputed** and goes to review.
- The feedback categories are a closed set (bug, idea, confusing, wrong, praise); labels are translated but the keys stay the same for analytics (section 5.10).
- Use of feedback for AI evaluation depends on a consent setting (`US-J06`).

4.10 Epic J: notifications, activity, settings and help

Epic purpose: the supporting features that make the workspace usable day to day: notifications, an activity feed, workspace settings, a guided tour and help, global search, and consent records.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-J01`	Must	Notifications in the app and by email digest, so users act when something waits on them.	Events: **invitation, step waiting on me, document ready, feedback answered, announcement**; **per-user preferences**; **daily or weekly digest**.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-J02`	Should	An activity feed showing what the team and the AI changed.	**Facts confirmed, documents created, use cases added, workflows started and completed**, with actor and time.
`US-J03`	Must	Workspace settings so the company controls its data and team.	**Name, logo, default language, members, Coordinator access, sharing with the Operator, data retention, export of all data (documents, profile, chats as JSON and Markdown in a ZIP) and a deletion request**.
`US-J04`	Must	A guided tour and help so a new user is productive on day one.	**First-run tour, help articles in both languages, contextual tips, an onboarding checklist on home that ticks itself off**.
`US-J05`	Should	Search across everything in the workspace.	**Profile, chats, documents, sources and use cases** searchable with **filters by type and date**.
`US-J06`	Must	The platform records consent so personal data and feedback are used lawfully.	**Terms and privacy notice accepted at first sign-in**; consent to **Operator visibility** and to **feedback use** are explicit settings with their **change history**.

What matters most

- Consent is a real feature: terms and privacy acceptance at first sign-in, plus two separate switchable consents (Operator visibility, feedback use), each with history. This links to `US-C08` (sharing) and `US-I06` (feedback use).
- Data export is a ZIP containing documents, profile and chats as JSON and Markdown (`US-J03`).
- v1 includes only `US-J01`, `J03`, `J04`, `J06` (section 11). `US-J02` and `US-J05` are Should and fall into v1.1.

4.11 Epic K: Japanese first, English equal

Epic purpose: brief item 9. Member companies work in Japanese; the Operator and Atomic Loops work in both languages. **Every element of the product** (navigation, dashboards, chats, generated profile, documents) exists in Japanese and in English, and a user must never meet an untranslated screen. `NFR-13` states the platform-level rule; these stories make it testable.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-K01`	Must	Every interface element exists in both languages.	**Navigation, buttons, forms, tables, charts, dashboards, emails, error messages and help** exist in both; **Japanese is the default for company workspaces**; **language toggle on every page**; an **automated test fails the build on any missing string**; **screenshots of every page in both languages reviewed before release**.
`US-K02`	Must	The AI replies in the user's language and switches when asked.	Replies follow the user's language; a chat can be switched to the other language; a **mixed-language message** (Japanese question with English product names) is handled; a **single reply in the other language** on request.
`US-K03`	Must	Every generated output is in the workspace language and available in the other language on demand.	**Profile, facts, documents, charters, recommendations and the road map** generated in the workspace language; a **one-click version in the other language**; the **workspace glossary and Appendix F** keep terminology consistent.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-K04`	Must	The AI reads Japanese websites and documents correctly.	**Japanese encodings, vertical text in PDFs where feasible and Japanese OCR** handled; facts are **stored in the source language** and shown with a translation labelled **"translated"** when the viewer's language differs.
`US-K05`	Must	The library and explainer content are bilingual by construction.	Every use case, stage and explainer has **Japanese and English fields**; **publishing is blocked until both exist**; the **Operator's reviewer approves the Japanese wording** (**`OQ-10`**).
`US-K06`	Must	Dashboards and charts are labelled in the user's language with local formats.	**Axis labels, legends, statuses and tooltips in both languages**; numbers, dates and currency formatted **per locale**, yen formatted per locale; **ISO dates in admin exports**.
`US-K07`	Should	Search and retrieval work across languages.	A query in either language retrieves sources in both; **citations show the source's original language**.
`US-K08`	Could	Voice works in both languages without choosing first.	**Language detected per utterance**; transcript stored in the spoken language; spoken replies in the user's language (**v1.1**).
`US-K09`	Must	Typing and reading behave correctly for Japanese.	**IME composition** handled in every field, including the chat composer; **Japanese line-breaking rules**; **fonts with full JIS coverage** on web and in exports.

ID	Priority	Plain meaning	Exact criteria from the SRS
`US-K10`	Should	Super Admin or Coordinator can read a company's shared content in their own language.	Shared sections and feedback shown in the viewer's language with a **"translated by AI"** marker where the original differs; **exports keep the original language** unless a translation is requested.
`US-K11`	Must	Notifications and emails arrive in the user's language.	**Every template exists in both languages**; the **recipient's language setting decides**; the **invitation email follows the workspace default until the user chooses**.

What matters most

- Bilingual is built into data (Japanese and English fields on library items, publishing blocked unless both exist) and into process (the build fails on any missing string).
- Japanese input method (IME) handling matters: pressing Enter to confirm a Japanese character conversion must not send a chat message by accident (`US-K09` , also section 5.10).
- Facts are stored in the language of the source; translations are labelled as translations.
- The AI's language rule: reply in the user's language, handle mixed messages, offer one reply in the other language on request.
- Appendix F lists the terms that need one agreed Japanese equivalent each; these are decided with the Operator's reviewers (`OQ-10`) and stored in a localisation file, not in the SRS.

SRS Section 5: Page examples

What this section does: it shows text wireframes (layout sketches) of the key pages, so layout and components are fixed before visual design. Labels are shown in English for readability; in the product they are Japanese. The sample company, **Sample Co. A**, is fictional: a **48-person precision machining company** making **hydraulic valve bodies and manifolds** for construction-equipment makers. All figures in the samples are illustrative.

The wireframes themselves are ASCII drawings in the SRS. Below, each one is described in words, with every component, state and sample value that the SRS gives.

5.1 Admin console: companies list and create company

Stories: `US-A01`, `US-B01`. **Route:** `/admin/companies`.

Companies list layout

- Top bar: `[Operator]` name, "Admin console", search, a bell with a notification count (sample: 3), a `JA | EN` language toggle, user menu.
- Left rail: Dashboard, Companies (selected), Users, Library, Templates, Feedback, Copilot, Audit log, Settings.
- Main area: heading "Companies (23)", a filter dropdown ("All"), and a **[+ Create]** button.
- Table columns: **Company, Status, Profile complete, Use cases, Last active**.
- Sample rows: Sample Co. A, Active, 78%, 4 use cases, 2 h ago; Sample Co. B, Onboarding, 35%, 0, 1 d; Sample Co. C, Invited, no data; Sample Co. D, Dormant, 91%, 7, 34 d.
- Footer: **Export CSV** and pagination.

Create company dialog fields

Field	Type
Company name (JA)	text
Company name (EN)	text
Industry	dropdown (sample: Metal processing)
Size band	dropdown (sample: 21 to 50 employees)
Default language	radio: Japanese (default shown) / English
Company Admin	name and email
Coordinator	dropdown (sample: none)
Buttons	Cancel, **Create and invite**

States: an empty list shows a "create your first company" prompt; row actions are **open, resend invitation, deactivate**; a **Dormant** badge appears after 30 days without activity.

5.2 Onboarding wizard

Stories: `US-C01` to `US-C03`. Shown to the Company Admin on first entry.

- **Four steps:** Language, Website, Facts, Review. Step 2 of 4 ("Website and products") is drawn.
- **Step 2 fields:** website URL(s) with "+ Add another"; a free-text box to describe products in your own words (sample text: "We machine hydraulic valve bodies and manifolds for construction-equipment makers, lot sizes 50 to 500 ..."); buttons **Upload catalogue (PDF, XLSX)**, **Add photos**, **Speak instead**.
- **"Products found so far"** list that fills as the AI works (sample: Hydraulic valve bodies, Manifolds, Pump housings, with a **+ Add** button).
- A reassurance note: the AI will read your website and catalogue, then ask a few questions; nothing is shared outside your workspace.
- Buttons: **Back** and **Analyse ->**.

Analysing state (progress panel)

- Checklist with live progress: read 38 pages from the website; found 3 product families, 2 certifications, 1 plant; analysing products and potential (in progress); preparing your profile (pending).
- A "Found so far" list with sources, for example `ISO 9001 (sourced: /company/quality)` and `5-axis machining centres (sourced: /equipment)`.

States: a site that cannot be fetched shows the reason and a "describe your company instead" path; **the analysis can run in the background while the user continues to step 3**; the wizard resumes where it stopped.

5.3 Company profile

Stories: `US-C04` to `US-C08`. **Route:** `/w/{company}/profile`.

Three-column layout

- **Left:** section menu: Overview, Products, Markets, Capabilities, Supply chain, Competition, Potential, Open questions, plus History.
- **Centre:** the selected section's facts. Header shows "Completeness 62%" and an **Export** menu. A "Visible to Operator" tick shows whether that section is shared.
- **Right:** "Gaps and next steps" (sample: Which three customers bring most revenue? leading to the Markets workflow; Capacity utilisation; How inspection results are recorded), an **Ask the copilot** button, and "Feedback on this section" Yes / No.

Each fact shows its provenance tag and controls

- Sample facts: "48 employees, Miyazaki" **[Stated]**; "Founded 1987" **[Sourced >]** (click opens the source); "Likely tier-2 supplier to two or three OEMs" **[Inferred, 70%]** with **Confirm / Edit / Reject** buttons.
- The Products section shows each product with function, customers and potential. Potential items are Inferred and have a **"Why?"** control that shows the facts and sources the inference rests on.

Sample AI wording for the Potential section (as the AI would phrase it): the 5-axis capability and OEM quality system suggest two adjacent markets (thermal-management manifolds for electric construction equipment; valve blocks for agricultural machinery made in Kyushu), both inferred from the website and public OEM announcements, with a request to confirm whether the company has quoted either.

States: a fact under review shows a **dotted border**; a **Disputed** fact shows the reporter's note; the **History** tab lists every change with before and after values and who made it.

5.4 Discovery workflow

Stories: `US-D02` to `US-D05`. **Route:** `/w/{company}/workflows/{id}`. The example is "Market and product-market fit" at step 2 of 5.

Layout: steps on the left, the workflow's own chat on the right. Header shows "Progress 2 of 5".

Step	Content	Status in the sample
1 Customers and markets	8 questions, answered by the sales manager	Done
2 Upload price list and recent quotes	Upload button and "I don't have this" option; owner: Admin	In progress
3 AI analysis	segments, win and loss patterns	Waits on step 2
4 Review hypotheses	you approve or edit	Not started
5 Output: Market and PMF report (document)	a generated document	Not started

Below the steps: **Dependency view**, **Pause**, **Assign step** buttons. The chat on the right shows the AI noting an answer as a **Stated fact, pending review**, with a text and hold-to-talk input.

Sample question set for step 1 (generated from profile gaps, asked a few at a time and adapted to answers):

1. Which three customers accounted for most of your revenue last year, and roughly what share each?
2. Which product family has grown fastest in the last two years, and what drove it?
3. Have you lost a quote in the last twelve months? To whom and why?
4. What is a typical lot size and lead time for valve bodies, from order to dispatch?
5. Who decides to buy at your customers: purchasing, engineering or both?
6. Which customers assemble outside Japan, and do they ask you to supply there?
7. Is there a product you have been asked for and declined? Why?
8. How are inspection results recorded today: paper, spreadsheet or a system?

Why the last question is there on purpose: a market workflow also **seeds** the Manufacturing and AI readiness workflow, so the AI learns about the plant while asking about the market.

5.5 Chat

Stories: `US-E01` to `US-E07`. **Route:** `/w/{company}/chats/{id}`.

Three-column layout

- **Left:** chat list with **+ New**, a selected chat (sample: "Export to Southeast Asia?"), other chats (Quality records, Grant letter), **Pinned** (Weekly plan) and **Archived** groups.
- **Centre:** the conversation. The sample AI reply lists three points with citation markers (`[1]` for profile/source, `[web >]` for a web source) and ends with an **"AI assumption, verify"** note on a landed-cost estimate (about +12%). Under the reply: **Copy, Save to KB, -> Document, -> Workflow, -> Profile, Helpful?**. At the bottom: composer "Ask, or hold to talk ...", **Mic** and **Attach** buttons.
- **Right (Context panel):** tick boxes for Profile sections (Overview, Products, Markets), Sources (2, e.g. price-list-2026 and iso-certificate), Workflow (Market and PMF, unticked in the sample), the linked workflow's step status, an expandable **summary of earlier messages**, and "Feedback on this chat".

States: streaming reply with a **stop control**; a reply given **without the profile carries a notice**; **voice mode shows the live transcript and a "send" confirmation**; the **token-budget notice appears above the composer at 80%**.

5.6 Knowledge base and document studio

Stories: `US-F03` to `US-F09`. **Routes:** `/w/{company}/knowledge` and `/w/{company}/knowledge/documents/{id}`.

Generate tab

- Tabs: **Sources (14)**, **Documents (6)**, **Generate** (selected).
- "What do you need?": a **Template** dropdown (sample: Business plan) **or** a free-text description (sample: a three-year business plan for a bank loan to add a second 5-axis machining centre).
- "Use:" tick boxes for context: Company profile, price-list-2026.xlsx, Workflow: Market, iso-certificate.pdf.
- **Language** (Japanese / English) and **Length** (sample: about 8 pages).
- Buttons: **Show outline first** and **Generate ->**.

Document editor

- Header: title with version ("Business plan (draft v2)"), **Versions**, **Share**, **Export: DOCX PDF**.
- Left: the document text with inline source markers (`[1]` , `[2]`) and **[AI assumption, verify]** tags.
- Right panel: **AI actions on selection** (Rewrite, Shorten, Expand, Translate JA/EN, Insert profile fact, Table from sources), **Sources used** (numbered list), **Comments (2)**, "Feedback on this document".

States: a source still processing is **greyed in the Use list**; a **failed generation keeps the request so it can be retried**; **"Save as source"** appears once a document is marked final.

5.7 AI use-case explorer

Stories: `US-H02` to `US-H07`. **Route:** `/w/{company}/ai-use-cases`.

Tabs: Method, Factory map (selected), Library, Mine (4), Road map.

Factory map: the seven stage boxes in flow order: Inward, Warehouse, Kitting, Assembly, Final inspection, Packaging, Logistics, with a "Planning and supply chain" lane above, and Quality, Maintenance, R&D and product discovery, and Sales and product-market fit lanes

below. **A star (*) marks stages indicated by the profile** (in the sample, Assembly).

Stage detail panel (sample: Assembly, "machining cells, in your case"): typical operations (operator, line leader, in-process inspector), typical systems (MES, ERP production, OEE tracking, torque tools), and use-case cards. Each card shows: title; entry point, horizon and level (for example "software only · H1 · L1"); a **"Why:"** line tied to a Stated fact; and a **[+ Add]** button. Two sample cards: Digital in-process inspection records (software only · H1 · L1) and Cycle-time monitoring per station (camera based · H2 · L2).

"My use case" charter (made with the wizard) is a one-page sheet with these fields: **Stage, Entry point, Horizon, Level needed, Problem, Change, Data, Systems, KPI, Creates** (what future data or capability this enables), and **Readiness** shown as four dot ratings (Data, Process, People, Systems) with a **[Check]** button; actions: **Edit, Place on road map, Make a charter document, Helpful?**. The sample use case is "Digital in-process inspection records" (Stage: Assembly (machining cells); Entry point: software only; Horizon: H1 quick win; Level needed: L1 (assisted); KPI: first-pass yield known the same shift, today two days).

Road map board: a 2x2 with vertical axis **Business impact** (low to high) and horizontal axis **Readiness to deploy** (low to high). Quadrants: top-left **Strategic bets**, top-right **Quick wins**, bottom-left **Later**, bottom-right **Momentum builders**. Under it, a **programme view**: H1 quick wins to H2 strategic bets to H3 disruptive, with dependencies drawn as "needs the data created by".

5.8 Portfolio dashboard

Story: `US-B03`. **Route:** `/admin`. The Super Admin's first screen each morning; aggregated and respecting sharing choices. Figures are sample data.

- **Top tiles:** Companies 23 (Invited 3); Active in 7 days 14 (61%); Profiles approved 16; Feedback positive 84% of 212 ratings. A period selector ("This month") and an **Export** button.
- **Onboarding funnel (sample):** Invited 23, Accepted 20, Profile 16, First workflow done 11.
- **Most-chosen use cases (sample):** 1 Digital inspection records; 2 Document OCR at receipt; 3 Daily plan digitised.
- **Industries (sample):** Metal processing 8, Food 5, Electronics 4, Plastics 3, other 3.
- **Needs attention:** 2 dormant companies, 5 feedback items unanswered.

5.9 Feedback widget

Story: `US-I02`. A small panel on every page: choose one of **Something is wrong, Idea, Confusing, Bug, Praise**; free text ("Tell us more …"); **Attach screenshot**; and the auto-captured context line (sample: "Page: /profile, JA"); **Send**. The per-output rating controls in 5.3 to 5.7 feed the same inbox.

5.10 Localisation of the pages

Every page ships in **both languages**. All strings are **externalised resources keyed by element** (the standard i18n approach). The SRS gives the English strings for the shell and chat page as the starting resource; the Japanese equivalents are produced during the build, approved by the Operator's reviewers (`OQ-10`), and live in a localisation file, not in the SRS.

Element	English strings	Developer note
Workspace navigation	Home, Company profile, Workflows, Chats, Knowledge base, Business planning, AI use cases, Feedback, Settings	Japanese labels are often wider; the rail reserves width and wraps to two lines where needed.
Top bar	Search, Notifications, Language, Ask the copilot	The language control shows each language in its own name.
Chat list	New chat, Pinned, Archived	Sort order and search must handle both scripts.
Composer	Ask, or hold to talk; Attach; Microphone	Placeholder and send action must survive IME composition (`US-K09`).
Context panel	Context, Profile, Sources, Linked workflow, Summary of earlier messages	The summary is generated in the user's language.
Message actions	Copy, Save to knowledge base, Turn into document, Turn into workflow, Propose to profile, Helpful?	Icon plus label; labels collapse into a menu on phones.

Element	English strings	Developer note
Provenance labels	Stated, Sourced, Inferred, AI assumption (verify)	One agreed term each, used identically in the interface, the AI's prose and exports (Appendix F).
Fact controls	Confirm, Edit, Reject, Why?	Short verbs; keep button widths flexible.
Profile	Completeness, Visible to Operator, Gaps and next steps, History	"Operator" is replaced by the organisation's name in both languages.
Workflow	Progress, Done, In progress, Waits on, Assign step, Pause, Dependency view	Status words are a closed set shared with the admin console.
Document studio	Sources, Documents, Generate, Show outline first, Versions, Export	Format names (DOCX, PDF, Markdown) are not translated.
Use-case explorer	Method, Factory map, Library, Mine, Road map, Recommended for you, Add	Stage names follow the agreed terms in Appendix F.
Road map quadrants	Quick wins, Strategic bets, Momentum builders, Later	Agreed terms needed; these also appear in exported documents.
Admin console	Portfolio dashboard, Companies, Create and invite, Feedback inbox, Audit log	The admin console defaults to the Operator's language.
Feedback widget	Something is wrong, Idea, Confusing, Bug, Praise, Tell us more, Send	Categories are a closed set used in analytics; translate the label, keep the key.

Sample bilingual chat reply: a Japanese workspace sees the reply from 5.5 in Japanese by default, generated by the same reply path from the same facts and citations. The English version reads: three things matter: (1) customers are construction-equipment OEMs [1], several assembling in Thailand [web]; (2) the ISO 9001 certification meets their local sourcing requirement [2]; (3) landed cost is estimated about 12% higher (AI assumption, verify).

What matters most (whole of section 5)

- The wireframes fix layout and components, not visual design.
- Three-column patterns repeat: list or menu on the left, content in the centre, context or gaps panel on the right.
- The **provenance tag, confidence value and Confirm / Edit / Reject controls** are the signature UI of the profile; the **context chips and message actions** are the signature UI of chat.
- In the feedback widget the wireframe's wording is "Something is wrong" while the story `US-I02` calls the category "wrong". The key stays the same; the label is what gets translated.
- Sample numbers (23 companies, 62% completeness, 84% positive, and so on) are illustrative data only.

SRS Section 6: AI agent specification

What this section does: it defines how the AI works internally. This is the section a developer building the chatbot needs most. It says the AI is **one orchestrating agent** that calls a small set of **specialised skills** and **tools**. Every call carries the company's **verified facts first**, cites what it used, and leaves a trace the user can inspect.

6.1 Skills (what the agent can do)

Skill	What it does	Where it appears	Main inputs	Outputs
Company analyst	Reads the website and product material, extracts facts, analyses products and potential	Onboarding, profile, re-analysis	URLs, uploads, basic facts	Proposed facts with provenance and confidence; product entries; gap list

Skill	What it does	Where it appears	Main inputs	Outputs
Interviewer	Generates adaptive question sets to close gaps and interprets answers into facts	Workflows, chats	Profile gaps, question banks, prior answers	Question sets; Stated facts; follow-up questions
Workflow planner	Proposes and drafts workflows from gaps, templates or a goal; proposes what comes next on completion	Workflows, chats	Profile, templates, goal text	Workflow definitions; next-workflow proposals
Document writer	Drafts outlines and documents from a prompt, a template and retrieved sources; runs the inline edit actions	Document studio, chats	Prompt, template, selected context	Outline; draft with citations and assumption markers
Use-case matcher and assessor	Ranks library use cases against the profile, drafts charters, classifies level and horizon, runs readiness checks	Use-case explorer	Profile, library, wizard answers	Ranked recommendations with rationale; charters; readiness ratings

Skill	What it does	Where it appears	Main inputs	Outputs
Planning coach (v1.1)	Pre-fills canvases, challenges assumptions, derives actions	Planning	Profile, sources, canvases	Canvas cells with provenance; critiques; tasks
Admin copilot (v1.1)	Turns natural-language admin requests into confirmed actions and answers portfolio questions	Admin console	Request text, aggregated data	Proposed action for confirmation; answers

6.2 Tools (what the agent can call; names exactly as specified)

Tool	Purpose	Side effect	Confirmation
`fetch_site(url, limits)`	Crawl public pages within robots.txt, depth and page limits; return text with page URLs	None	Not needed
`parse_document(file)`	Extract text, tables and images from uploads, with OCR when needed	None	Not needed

Tool	Purpose	Side effect	Confirmation
`search_knowledge(query, scope)`	Hybrid keyword and semantic retrieval over the workspace's sources, documents and chats, returning chunk citations	None	Not needed
`read_profile(sections)`	Return confirmed facts first, then proposed facts flagged as unconfirmed	None	Not needed
`propose_fact(fact)`	Add a fact to the review queue	Writes to the queue	The review queue is the confirmation
`create_workflow`, `update_step`	Create a workflow or change a step's status	Writes	Yes, in the interface
`generate_document(request)`	Produce an outline or a draft into the document studio	Writes a draft	Not needed; the draft is editable
`match_use_cases(profile)`	Score library use cases against the profile	None	Not needed
`speech_to_text`, `text_to_speech`	Voice in and out, Japanese and English	None	Not needed

Tool	Purpose	Side effect	Confirmation
`web_search` (restricted)	Look up public facts about markets, standards or customers named in the profile; results cited	None	Not needed; results labelled **Sourced (web)**
`admin.create_company`, `admin.invite_user` (v1.1)	Admin Copilot actions	Writes, sends email	Yes, explicit confirmation showing the exact values

The pattern is simple: **read-only tools run freely; tools that write something visible or send something need a human confirmation**, except `propose_fact` (the review queue itself is the confirmation) and `generate_document` (the draft is editable, so it is safe).

6.3 The company fact ledger (the agent's memory)

The ledger is the AI's memory of a company. It is **the only store the profile, the matcher and the writer read from**, so one correction fixes everything that depends on it.

Fields: `id`, `workspace`, `section`, `key`, `value`, `unit and period`, `language`, `provenance` (Stated, Sourced, Inferred), `sources` (URL, document chunk, chat message or answer id), `confidence` from **0 to 1**, `status` (**Proposed, Confirmed, Rejected, Disputed, Superseded**), `created by` (user or skill), `timestamps`, `supersedes`.

Rules

- **Confirmed facts outrank Proposed facts; Rejected facts are never used.**
- An **Inferred** fact is shown with its label and its "why" trail; it becomes **Stated** only when a user confirms it.
- **Conflicts** (two values for one key) are surfaced in the review queue and **never resolved silently**.
- A number with **no source is an estimate** and is labelled as one.
- **Deleting a source** marks the facts drawn only from it as **Unsourced** and sends them back to review.

6.4 Context assembly for each model call (the order of the prompt)

Every call to the language model is built from these pieces, in this order:

1. The **policy and role prompt**, versioned, including the user's language and the no-side-effect rule.
2. **Workspace instructions** set by the Company Admin: tone, terminology, what to avoid.
3. **Confirmed profile facts** for the relevant sections, then **Proposed facts** flagged as unconfirmed.
4. **Retrieved chunks** from the knowledge base with citation handles, **scoped to this workspace only**.
5. **Workflow state** (current step, prior answers) when the chat is linked to a workflow.
6. **Conversation history**, with older turns summarised; the summary is visible to the user.
7. The **user's message and attachments**.

6.5 Guardrails (rules the AI must follow)

Guardrail	Meaning
Fetched and uploaded content is data	Instruction-like text inside it is kept out of the instruction channel, quoted as data and flagged on the source (`US-C10`).
No side effect without confirmation	Tools that write outside the draft space require an interface confirmation that shows the exact values.
No fabricated specifics	Numbers, names and dates must be Stated or Sourced; otherwise the output says it is an estimate or asks.
Scope	The AI gives business and manufacturing guidance; on **legal, tax and investment** matters it gives general information and recommends a professional.
Privacy	Personal data is minimised in prompts; names of people in uploads are **not extracted into the profile** unless a user adds them.
Isolation	Retrieval, memory and caches are partitioned **per workspace**; the **library and templates are the only shared content**.

Guardrail	Meaning
Safety	Content filters on input and output; a refusal explains what the AI cannot do and what it can.
Transparency	Every reply can show what it used (the **context chips**) and why it inferred what it inferred.

6.6 Quality and evaluation (release targets)

What is measured	How	Target for release
Fact extraction precision	A **golden set of 30 Japanese manufacturer websites**; extracted facts checked by a reviewer	**90%** of extracted facts correct; every source link resolves
Question relevance	Reviewer rating of generated question sets against the profile gaps	**4 of 5** on average
Citation coverage in documents	Share of sourced statements that carry a resolvable citation	**95%**
Assumption labelling	Share of unsourced numbers that carry the estimate marker	**100%** in the golden set
Recommendation fit	Reviewer rating of the top five recommendations per golden profile	**4 of 5** on average
User feedback	Positive rate on AI outputs from `US-I01`	**80%** after the first quarter
Injection resistance	A red-team set of crafted pages and files	**No behaviour change in any case**
Latency	Time to first token in chat; time to a generated profile	As in section 9

Prompts, templates and models are **versioned**. A change ships only after the evaluation set passes, and the previous version stays available for **rollback**.

6.7 Model and cost strategy

- **Provider-agnostic:** each skill is configured with a model tier (**quality or fast**) and can be switched without code changes; Japanese capability is a selection criterion.
- **Embeddings** are chosen for Japanese and English retrieval; a workspace's index is **rebuilt when the embedding model changes**.
- Website analyses and document extractions are **cached per source version**; repeated questions reuse retrieval results.
- **Budgets:** a monthly token budget per workspace (`US-E09`), per-call limits on context size; long outputs stream and can resume.
- **Every model call is logged** with references to its inputs (not their content), tokens, latency and cost, so quality and spend are traceable per feature.

What matters most

- One orchestrating agent, seven skills (five in v1, two in v1.1), a tool table of 11 rows covering 14 individual tools.
- The fact ledger is the single source of truth. The profile, the use-case matcher and the document writer all read from it. Build it first.
- Prompt order matters: policy, workspace instructions, confirmed facts, proposed facts (flagged), retrieved chunks, workflow state, history, user message.
- Provenance (Stated, Sourced, Inferred) plus confidence (0 to 1) plus status (Proposed, Confirmed, Rejected, Disputed, Superseded) are three separate fields on every fact.
- Retrieval must be hybrid (keyword and semantic) and partitioned per workspace.
- The evaluation set (30 Japanese manufacturer websites) and its targets are the quality gate for each release.

SRS Section 7: Data model

What this section does: it lists the data entities (the tables or collections of the database), their key fields and how they relate. The model has **one root per company, the workspace**, and keeps three kinds of things apart:

1. **What users and the AI know:** facts and sources.
2. **What they do:** workflows, chats, documents, use cases.
3. **What the platform records about it:** feedback, audit, consent.

Entity	Key fields	Relationships	Plain note
Organisation	Name, type (Operator or Atomic Loops), default language, branding, retention policy	Has staff Users; owns Workspaces	The two organisations that run the platform.
Workspace	Slug, names (JA, EN), industry, size band, status, default language, token budget, sharing settings, retention	Belongs to an Organisation; owns everything below	The root of every company's data. Every query is scoped by it.
User	Email, names, language, roles per scope, MFA, consents, notification preferences	Member of an Organisation or a Workspace; Coordinator grants	The SRS says roles are held "per scope" (the Operator organisation, or one specific workspace), and that Coordinator access is granted per workspace.
Invitation	Email, role, workspace, token, expiry, status	Created by a User	Backs the `/invite/{token}` route.
Profile section	Key, completeness, shared with Operator (yes or no)	Holds Facts	The eight profile sections; sharing is set per section.
Fact	Section, key, value, unit and period, provenance, confidence, status, sources, created by, supersedes	Belongs to a Profile section; cites Sources, Messages or Answers	The fact ledger of section 6.3.
Product	Name, family, function, materials or process, customers, differentiators, potential	Belongs to a Workspace; has Facts	One entry per product or product family.

Entity	Key fields	Relationships	Plain note
Source	Type (upload, URL, saved document, saved chat), file, extracted text, status, tags, chunks and embeddings	Belongs to a Workspace; cited by Facts, Messages and Documents	Knowledge-base item; chunks and embeddings power retrieval.
Workflow	Template and version, title, goal, status, created by (AI or user)	Has Steps, one Chat and Outputs	One chat per workflow.
Step	Type, order, status, owner, due date, question set, answers, output reference	Belongs to a Workflow	Five types, five statuses (`US-D02`).
Answer	Question, value, mode (text, voice, upload), answered by	Belongs to a Step; produces Facts	Each answer can create a Stated fact.
Chat	Title, linked workflow or document, pinned, archived, summary	Has Messages	Summary supports long chats (`US-E10`).
Message	Role (user or AI), content, attachments, citations, context used, feedback	Belongs to a Chat	Stores which context was used, for transparency.
Document	Title, template, language, status, versions, sources used, assumption markers	Belongs to a Workspace; may become a Source	Versions and assumption markers are stored.
Template	Type (workflow, document, question bank), version, language fields, published	Owned by the Platform Admin or a Workspace	Platform-wide or company-owned.

Entity	Key fields	Relationships	Plain note
Library use case	Stage, titles and texts (JA, EN), entry point, horizon, autonomy level, data needed, systems, prerequisites, status, version	Published by the Platform Admin	Shared content, not tenant data.
Company use case	Library reference or custom, charter fields, readiness scores, quadrant, horizon, dependencies	Belongs to a Workspace	May point to a library item or be fully custom.
Assessment request	Status, pre-filled snapshot, contact	Belongs to a Workspace; routed to the Organisation	For `US-H09` (v1.1).
Feedback	Type, rating, reason, text, screenshot, page, target reference, status, reply	By a User; about a Message, Fact, Document, Use case or Page	Target reference lets feedback point at any object.
Notification	Event, recipient, channel, read	To a User	Channels: in-app and email.
Audit entry	Actor, action, target, workspace, time, before and after	Immutable	Never edited or deleted.
Consent	User, kind (terms, Operator visibility, feedback use), value, time	By a User	History of consent changes.

What matters most

- The **Workspace** is the root. Everything company-owned hangs off it; this is how tenant isolation works in the data model.

- **Library use cases and templates are shared** across companies; they are the only shared content.
- A Fact can cite **Sources, Messages or Answers**, so every fact can be traced back.
- Feedback uses a flexible target reference so it can be about a message, fact, document, use case or page.
- The audit entry stores **before and after** values and is immutable.

SRS Section 8: External interfaces and integrations

What this section does: it lists everything the platform depends on outside its own code, so provider decisions (section 12) can be made against it.

Interface	Purpose	Release	Notes
Web application	All user interaction, responsive, Japanese and English	v1	Accessibility per section 9; works in phone browsers
Large-language-model API	All generation, extraction and matching	v1	Provider-agnostic adapter; approved processing region; no training on customer data
Embedding and vector index	Retrieval over sources	v1	Partitioned per workspace
Web fetcher	Website analysis and restricted web search	v1	robots.txt and rate limits; allow-list for search
Document processing	Text, table and image extraction with Japanese OCR	v1	Runs inside the platform's own environment
Speech-to-text and text-to-speech	Voice in chats and workflows	v1.1	Japanese and English; streaming
Email	Invitations, notifications, digests	v1	Transactional provider; Japanese templates
Object storage	Uploads, exports, screenshots	v1	Encrypted; region per `OQ-03`

Interface	Purpose	Release	Notes
Export	DOCX, PDF, Markdown, CSV, ZIP	v1	Generated server-side
Single sign-on	Operator staff sign-in with their identity provider	v2	SAML or OIDC
Public API	Read access for the Operator's own reporting	v2	Scoped tokens
Member companies' ERP, MES and machines	Not connected	None	The platform plans use cases; execution in the plant is outside scope

What matters most

- For v1 you need: the web app, an LLM API (via a provider-agnostic adapter), embeddings plus a vector index (partitioned per workspace), a web fetcher, document processing with Japanese OCR, email and object storage, and server-side export.
- Voice services (speech-to-text and text-to-speech) are v1.1. Single sign-on and the public API are v2.
- Document processing is meant to run **inside the platform's own environment** (not sent to an outside service).
- Web search is restricted by an **allow-list**.

SRS Section 9: Non-functional requirements (NFR-01 to NFR-20)

What this section does: it sets quality targets (speed, security, privacy, availability and so on) rather than features. The platform handles confidential material for many competing companies at once, so **isolation, privacy and provenance carry the same weight as performance**.

ID	Category	Exact requirement	Plain meaning
`NFR-01`	Performance	Chat first token within **2 seconds at p95**; page loads within **2 seconds at p95** on standard broadband in Japan.	95 out of 100 requests must meet these times. Both chat start and page load.
`NFR-02`	Performance	Website analysis completes within **5 minutes for sites up to 200 pages**, with live progress; a document of about **2,000 words generates within 90 seconds**.	Two timing targets: site crawl and document generation.
`NFR-03`	Capacity	v1 sized for **500 workspaces, 5,000 users and 200 concurrent users** without degradation; scales horizontally beyond that.	Sizing target for v1; the design must be able to add servers.
`NFR-04`	Availability	**99.5% monthly availability** for v1, measured on the web application and chat; maintenance windows announced **48 hours ahead**.	About 3.6 hours of allowed downtime in a 30-day month (this arithmetic is explanation, not in the SRS).
`NFR-05`	Backup and recovery	**Daily backups; recovery point 24 hours; recovery time 8 hours; restore tested quarterly.**	At most 24 hours of data can be lost; service must be back within 8 hours of a failure.
`NFR-06`	Security	**OWASP ASVS level 2; TLS 1.2 or higher in transit; encryption at rest; secrets in a managed vault; dependency scanning; penetration test before launch and yearly.**	The security baseline.

ID	Category	Exact requirement	Plain meaning
`NFR-07`	Access control	**Role-based access as in section 3.3, enforced server-side; MFA for admin roles; sessions expire after 12 hours idle.**	Permissions are checked on the server, never only in the interface.
`NFR-08`	Tenant isolation	**Every data access scoped by workspace; automated cross-tenant tests in the delivery pipeline; retrieval and caches partitioned per workspace.**	Same rule as `US-A07`, made a quality requirement.
`NFR-09`	Audit	**Immutable log of admin actions and of Operator access to company data, retained 1 year; user-visible history on facts and documents.**	Two kinds of history: the admin audit log and the visible history on facts and documents.
`NFR-10`	Privacy	**APPI compliance: purpose of use stated, consent recorded, data-subject requests handled within 30 days, personal data minimised in prompts and logs.**	Japan's privacy law.
`NFR-11`	Data residency	**Customer data stored in a cloud region in Japan; model processing region per OQ-04; data-processing agreements with every provider; no training on customer data.**	Data at rest is in Japan; where the model processes it is still an open decision.

ID	Category	Exact requirement	Plain meaning
NFR-12	Retention and deletion	**Workspace retention configurable, default 24 months after deactivation; deletion requests completed within 30 days with backups expiring after; export offered before deletion.**	How long data is kept and how it is removed.
NFR-13	Internationalisation	**Japanese by default and English, switchable per user; all content, emails and AI replies in the user's language; date, number and currency formats per locale; Japanese typography and input tested.**	The bilingual rule at platform level (Epic K).
NFR-14	Accessibility	**WCAG 2.1 AA and JIS X 8341-3; keyboard operable; screen-reader labels; voice as an alternative input.**	Accessibility standards.
NFR-15	Usability	**A Company Admin reaches an approved profile in under one hour of their own time with no training session; phone layouts for chat, workflows and feedback.**	Measurable ease-of-use goal.
NFR-16	AI reliability	**Provenance on every fact; citation and assumption markers in documents; the evaluation thresholds in section 6.6 met before each release; graceful fallback when a provider is down.**	The AI must keep working or degrade safely if the model provider fails.

ID	Category	Exact requirement	Plain meaning
`NFR-17`	Observability	**Structured logs, traces and metrics; model-call logging without content; alerts on error rate, latency and budget.**	Monitoring; do not log the content of model calls.
`NFR-18`	Maintainability	**Prompts, templates and library content versioned and editable without a deploy; feature flags per workspace; automated tests on every change.**	Content changes without redeploying; features switched on per company.
`NFR-19`	Browser support	**Current and previous major versions of Chrome, Edge, Safari and Firefox; Safari on iOS and Chrome on Android.**	Same list as section 2.4.
`NFR-20`	Cost control	**Per-workspace token budgets; per-call context limits; a monthly cost report per feature for the Platform Admin.**	Pairs with `US-E09`.

What matters most

- The numbers to keep handy: 2 s (p95), 5 min / 200 pages, 90 s / 2,000 words, 500 / 5,000 / 200 (workspaces / users / concurrent), 99.5%, 48 h notice, RPO 24 h and RTO 8 h, 12 h idle, 30 days (privacy requests and deletion), 24 months default retention, 1 year audit retention.
- Two points that can be tested early in a POC: tenant isolation (`NFR-08`) and logging without content (`NFR-17`).
- `NFR-18` requires feature flags per workspace and content editable without a deploy; plan the admin console accordingly.

SRS Section 10: Success metrics

What this section does: it defines how the platform will be judged. Targets are for the **first full quarter after v1** and **will be reset with real data**. The aim: companies keep coming back and the Operator spends minutes, not meetings, on each onboarding.

Metric	Definition	Target after the first full quarter
Time to approved profile	From invitation sent to profile approved by the Company Admin	**Median under 2 days**
Profile completeness	Average completeness score across active workspaces	**70% within 2 weeks of onboarding**
Weekly active companies	Workspaces with at least one user session in the week	**60% of active workspaces**
Workflows completed	Completed workflows per active workspace per month	**2**
Documents generated	Documents created per active workspace per month	**3**
Use cases defined	Company use cases on the road map per active workspace	**3**
Feedback positive rate	Share of thumbs-up on rated AI outputs	**80%**
Feedback answered	Feedback items addressed within 5 working days	**90%**
Assessment requests	Requests raised through the platform	Tracked; no target in v1
Operator effort	Super Admin time per onboarding	**Under 15 minutes**

What matters most

- Metrics need data to be captured: invitation sent time, profile approval time, sessions per week, workflows completed, documents generated, use cases on the road map, feedback and reply times, Super Admin time per onboarding.

- Do not confuse this table's 80% feedback positive rate (first quarter) with the pilot gate in section 11 (70% positive in the pilot).

SRS Section 11: Release scope

What this section does: it says exactly what ships when. Three releases, each **gated on evidence from real companies rather than on a date**: **v1** is the smallest product the Operator can onboard companies with; **v1.1** adds voice, planning and the lite assessment; **v2** opens the platform to programmes, identity providers and other organisations.

Release	Scope (exact)	Exit gate (exact)
v1	Epic A; Epic B (`US-B01` to `B04` , `B06` , `B07`); Epic C; Epic D (`US-D01` to `D05` , `D08`); Epic E except voice; Epic F (`US-F01` to `F03` , `F05` to `F07` , `F09`); Epic H (`US-H01` to `H05` , `H07` , `H10`); Epic I (`US-I01` to `I06`); Epic J (`US-J01` , `J03` , `J04` , `J06`); Japanese and English; evaluation thresholds met	**Five pilot companies onboarded by the Operator; positive feedback rate of 70% in the pilot; security review passed**
v1.1	Voice (`US-E04`); planning canvases (Epic G); readiness check (`US-H06`); lite assessment and assessment request (`US-H08` , `H09`); Admin Copilot (`US-B08`); announcements (`US-B09`); workflow template editor (`US-B05`); the remaining Should and Could stories	**Pilot companies using voice and planning; the lite assessment produces a preliminary summary that Atomic Loops has reviewed**
v2	Programmes and connections (`US-B10`); single sign-on; public API; whole-document translation (`US-F10`); support for a second organisation (another prefecture or agency)	**The Operator confirms the programme-matching rules; a second organisation is signed**

What is deliberately not in any release: connecting to member companies' ERP, MES or machines, and running AI use cases in the plant. Those remain the work of an Atomic Assessment and its follow-on projects.

A story-by-story view of the v1 list

The table above lists v1 by range. Spelled out, the stories that are in v1 are:

Epic	In v1	Not in v1
A (access)	all: `US-A01` to `A10`	none
B (admin console)	`US-B01`, `B02`, `B03`, `B04`, `B06`, `B07`	`B05` and `B08`, `B09` (v1.1); `B10` (v2)
C (profile)	all: `US-C01` to `C10`	none
D (workflows)	`US-D01`, `D02`, `D03`, `D04`, `D05`, `D08`	`D06`, `D07`, `D09`
E (chats)	all except voice (`US-E04`)	`US-E04` (v1.1)
F (knowledge base)	`US-F01`, `F02`, `F03`, `F05`, `F06`, `F07`, `F09`	`F04`, `F08`; `F10` (v2)
G (planning)	none	all (v1.1)
H (use cases)	`US-H01`, `H02`, `H03`, `H04`, `H05`, `H07`, `H10`	`H06`, `H08`, `H09` (v1.1)
I (feedback)	`US-I01` to `I06`	`I07`
J (settings)	`US-J01`, `J03`, `J04`, `J06`	`J02`, `J05`
K (language)	the SRS text says "Japanese and English" is in v1; it does not list Epic K's individual stories in the v1 row	`US-K08` is explicitly v1.1

Inconsistencies in the SRS worth raising with the document owner

These are places where the SRS contradicts itself or leaves something unclear. They are the kind of thing a POC team should clarify early:

1. **"Should" stories and v1.** Section 1.6 says Must **and Should** are in v1. But section 11 leaves several Should stories out of the v1 list (for example `US-B05`, `D06`, `D07`, `D09`, `F04`, `F08`, `H06`, `I07`, `J02`, `J05`), while Epics A, C and E (with Should stories such as `US-A06`, `A08`, `C07`, `C09`, `E06`, `E08`) are included wholesale. Section 11's v1.1 row says "the remaining Should and Could stories", which supports the section 11 reading.
2. **`US-F10` (whole-document translation).** Priority is **Could**, which by section 1.6 means v1.1, but section 11 puts it in **v2**.

3. **`US-H06` (readiness check).** Priority **Should**, yet section 11 explicitly puts it in **v1.1**. The use-case wireframe in 5.7 already shows readiness dots and a [Check] button.
4. **Epic K.** Section 11's v1 row says "Japanese and English" but does not enumerate the K stories. Most are Must, so they should be treated as v1.
5. **Journey J1.** Section 3.4 lists J2 to J5 only; J1 is not written out in the text.
6. **Admin Copilot.** The wireframe left rail in 5.1 shows "Copilot" even though `US-B08` is v1.1.

SRS Section 12: Open questions and decisions needed

What this section does: it lists **fifteen decisions** that shape the build. **The first four block the architecture and should be settled with the Operator before design starts.** In the SRS each has an empty tick box (not yet decided).

ID	Question	Why it matters
`OQ-01`	Product name, Japanese name and branding: the Operator's brand with "built with Atomic Loops", or a joint brand?	Naming and look of the product. **Blocks architecture.**
`OQ-02`	Who is the legal operator of the service and the data controller under APPI: the Operator, with Atomic Loops as processor?	Legal responsibility for personal data. **Blocks architecture.**
`OQ-03`	Hosting provider and region for data at rest (a region in Japan is assumed).	Where data is stored. **Blocks architecture.**
`OQ-04`	Approved large-language-model provider and processing region, with confirmation that its terms exclude training on customer data.	Which AI model you can use. **Blocks architecture.**
`OQ-05`	Speech provider for Japanese voice (v1.1).	Voice feature.
`OQ-06`	Email sender identity: the Operator's domain or the platform's.	How invitation emails look and are authenticated.

ID	Question	Why it matters
`OQ-07`	Default sharing with the Operator: overview and products summary only (proposed), or more?	The default for `US-C08`.
`OQ-08`	Does the Coordinator role match how the Operator works today, and should it be in v1?	Whether Coordinator features stay in v1.
`OQ-09`	Expected number of companies in year one and the token budget per workspace.	Capacity and cost planning.
`OQ-10`	Japanese review of the library and explainer content: who at the Operator reviews it, and how often?	Required for bilingual publishing (`US-K05`).
`OQ-11`	Is the lite assessment positioned as preparation for an Atomic Assessment, or as a stand-alone deliverable the Operator presents?	Positioning of v1.1 lite assessment.
`OQ-12`	Support model: who answers company questions in Japanese, and in which hours?	Support.
`OQ-13`	Sign-in: email and password with MFA (proposed), or the Operator's identity provider from v1?	Sign-in design.
`OQ-14`	Terms of use and privacy notice: drafted by whose counsel?	Legal content.
`OQ-15`	Next step: a clickable prototype of the onboarding wizard, the profile and the use-case explorer (sections 5.2, 5.3 and 5.7) before build, to test with two or three member companies?	Whether to prototype before build; note these three pages are the same ones that a POC would show.

What matters most

- The v1 exit gate: five pilot companies, 70% positive feedback in the pilot, security review passed.

- The SRS itself suggests prototyping the **onboarding wizard (5.2), profile (5.3) and use-case explorer (5.7)** first (`OQ-15`). That is a strong hint for what a POC should demonstrate.
- Until `OQ-03` and `OQ-04` are decided, the hosting region and the model provider are assumptions. For a POC, keep both behind adapters so they can be swapped.

SRS Section 13: Appendices

Six appendices (A to F) give the actual content that the product starts with: the use-case library, workflow templates, sample questions, document templates, the traceability table and the localisation glossary. Appendix content is data a developer will load into the system, so entries below are kept **exactly as in the SRS**.

Appendix A: use-case library seed

What it is: the starting content of the library, condensed from the Atomic Loops Factory Assessment Framework and the Atomic Assessment method. **Each row becomes a card** (`US-H03`). The library spans every stage and **grows after launch**; value ranges are added only where Atomic Loops has measured them and **never name a client**. There are **44 rows**.

How to read the columns

- **Stage or lane:** where in the factory flow it applies (or a supporting lane).
- **Entry point:** how data enters: *software only*, *camera based*, *machine IoT*.
- **Horizon:** H1 quick win, H2 strategic bet, H3 disruptive.
- **Level:** autonomy level L1 to L4 (names in Appendix F: L0 Manual, L1 Assisted, L2 Partial, L3 Conditional, L4 High, L5 Full).
- **Data it needs:** the input the use case depends on.

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Step 0, all stages	Factory intelligence dashboard: one source of truth with a view per role (leadership, production, quality, maintenance, R&D, supply chain, sales)	Software only	H1	L1	Records already in ERP and spreadsheets

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Inward	Document reading at the gate: invoice, delivery note and test certificates read and matched to the purchase order	Software only	H1	L1	Purchase orders, scanned documents
Inward	Material test certificate check against the specification library	Software only	H1	L1	Certificates, specification thresholds
Inward	Gate-to-receipt cycle time tracking by QR scan	Software only	H1	L1	Arrival and receipt timestamps
Inward	Supplier risk scoring from delivery and quality history	Software only	H2	L2	Delivery performance, rejection history
Warehouse	Part identification from a photo for unlabelled material	Camera based	H2	L2	Master product images
Warehouse	Put-away bin recommendation	Software only	H2	L2	Item dimensions, usage frequency, bin master
Warehouse	FIFO exception alerts and inventory anomaly detection	Software only	H1	L1	Lot dates, stock movements
Warehouse	Demand-driven replenishment ahead of a line shortage	Software only	H2	L3	Production schedule, consumption rates
Kitting	Kit completeness check before dispatch to the line	Camera based	H2	L2	Bill of kit, kit images
Kitting	Shortage prediction hours ahead of a kit stop	Software only	H2	L2	Stock, open orders, consumption history
Kitting	Lot-to-kit traceability	Software only	H1	L1	Lot numbers, kit numbers, serials
Assembly or machining	Digital in-process inspection records	Software only	H1	L1	Job, machine, operator, result
Assembly or machining	Cycle-time monitoring per station against takt	Camera based or machine IoT	H2	L2	Cycle start and end per unit
Assembly or machining	Pick-and-place and assembly verification	Camera based	H2	L2	Station images, work instructions
Assembly or machining	OEE and bottleneck finder	Machine IoT	H2	L2	Machine states, cycle logs, downtime reasons

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Assembly or machining	Andon root-cause suggestions	Software only	H2	L2	Andon history by station, shift and model
Assembly or machining	Predictive maintenance on critical machines	Machine IoT	H3	L3	Vibration, temperature and current history
Assembly or machining	Visual defect inspection against golden samples	Camera based	H2	L2	Labelled defect images
Final inspection	Mobile inspection app with enforced photo evidence	Software only	H1	L1	Checklists, serials
Final inspection	Automatic inspection report and certificate generation	Software only	H1	L1	Checklist results, test data
Final inspection	Non-conformance pattern analysis and corrective-action suggestions	Software only	H2	L2	NCR history, CAPA records
Final inspection	Dispatch hold linked to the inspection outcome	Software only	H1	L1	Inspection result, delivery order
Packaging	Label verification against master data	Software only	H1	L1	Label fields, ERP master
Packaging	Wrong-pack detection by weight or image	Camera based	H2	L2	Pack specifications, weights
Packaging	Serialised traceability from raw lot to customer order	Software only	H1	L1	Lot, serial, carton, order
Logistics	Load plan optimisation per truck	Software only	H2	L2	Carton dimensions, orders, departure times
Logistics	On-time delivery prediction and proactive customer alerts	Software only	H2	L2	GPS, route history, delivery windows
Logistics	Proof-of-delivery reading and validation	Software only	H1	L1	POD photos, delivery notes
Planning and supply chain	Daily production plan digitised with change history	Software only	H1	L1	Plan spreadsheets
Planning and supply chain	Finite-capacity scheduling on top of the ERP plan	Software only	H2	L3	Routings, capacities, orders
Planning and supply chain	Demand forecasting at warehouse and SKU level	Software only	H2	L2	Sales history, distributor data

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Planning and supply chain	Self-scheduling lines with planner approval of exceptions	Software only	H3	L4	All of the above, matured
Quality	Automated quality Pareto and first-pass-yield dashboard	Software only	H1	L1	Rejection records
Quality	Quality-drift warnings and root-cause suggestions	Software only	H2	L2	Inspection history linked to machine, batch and shift
Quality	Closed-loop quality with parameter correction	Machine IoT	H3	L4	Parameters, outcomes, drift history
Maintenance	Breakdown notification linked to production loss	Software only	H1	L1	Breakdown tickets, output records
Maintenance	Self-tuning machine parameters	Machine IoT	H3	L4	Parameter and outcome history
R&D and product discovery	Market and complaint signals as new-product triggers	Software only	H2	L2	Sales, complaints, market sources
R&D and product discovery	Bill-of-material and cell-change impact simulation	Software only	H2	L2	BOM, routings, cell data
R&D and product discovery	Product-market fit hypothesis tracking with evidence	Software only	H1	L1	Profile, sources, workflow answers
Sales and product-market fit	Must-sell recommendations per customer or outlet	Software only	H2	L2	Sales by customer and product
Sales and product-market fit	Sales pitch and role-play training built from product and application knowledge	Software only	H1	L1	Product documents
Sales and product-market fit	Competitor and review intelligence	Software only	H2	L2	Public sources

What matters most

- The library already follows the field structure of a library use case card: stage, title, entry point, horizon, level, data needed. Fields such as problem solved, how it works, systems touched, prerequisites and what good looks like (also required by `US-H03`)

are not in this seed table and must be written for each card.

- One use case, "Cycle-time monitoring per station against takt", has two entry points (camera based or machine IoT). The data model should allow more than one entry point value.
- Horizon and level generally rise together: H1 items are mostly L1, H2 mostly L2, H3 items are L3 or L4.
- The seed contains no value numbers (savings and so on); those are added later only if Atomic Loops has measured ranges.

Appendix B: workflow templates

What it is: each template is a **numbered sequence of steps** of the five types in `US-D02` (question set, upload request, AI analysis, review and approval, output document). Steps marked **(AI)** run **without the user**; the others need the company. Templates are **versioned by the Platform Admin** (`US-B05`). There are **eight templates, B1 to B8**.

B1, Company discovery (proposed to every new workspace)

1. Confirm the generated overview (review).
2. Question set: people and roles, plants and shifts, certifications, main customer types (about 10 questions).
3. Upload, optional: product catalogue, organisation chart, a recent company introduction.
4. (AI) Analysis: gaps, inferred market position, proposed products list.
5. Review the proposed facts.
6. Output: profile version 1 approved; the AI proposes the next workflows.

B2, Product deep-dive (one per main product family)

1. Question set: function, variants, materials and process, volumes and lot sizes, pricing position, who buys and why.
2. Upload, optional: drawings, specification sheet, photos.
3. (AI) Analysis: differentiators, adjacent applications, risks, what the product needs from the plant.
4. Review.
5. Output: a product one-pager; the product entry in the profile is updated.

B3, Market and product-market fit (the one drawn in section 5.4)

1. Question set: customers and markets.

2. Upload: price list and recent quotes.
3. (AI) Analysis: segments, win and loss patterns, demand signals.
4. Review the hypotheses.
5. Output: market and product-market fit report; hypotheses go to the planning board (v1.1).

B4, Manufacturing and AI readiness

1. Question set: lines, machines, shifts, systems in use.
2. Stage-by-stage capture: which stages exist and how the result is recorded at each (paper, spreadsheet, system, sensor).
3. Pain-point interview in the workflow's chat.
4. (AI) Analysis: recommended use cases, readiness gaps, the data each quick win would create.
5. Review.
6. Output: use-case candidates placed on the road map.

B5, Business plan

1. Goal and audience: bank, customer or a prefecture programme.
2. Question set: targets, planned investments, headcount, financing.
3. Context: profile, canvases and selected sources.
4. Outline approval.
5. (AI) Draft, then edit.
6. Output: business plan document.

B6, Export readiness

1. Target markets.
2. Question set: certifications, logistics, pricing, language, partners.
3. (AI) Analysis: gaps and the partners needed.
4. Output: export readiness memo.

B7, AI use-case charter

1. Wizard (`US-H05`).
2. Readiness check (`US-H06`, v1.1).
3. Output: charter document and placement on the road map.

B8, Lite assessment (v1.1)

1. KPI alignment questions.
2. Pain-point interview.
3. Capture per line or machine: the data lane, the human lane and the material lane.
4. Document checklist.
5. (AI) Analysis.
6. Output: preliminary readiness summary with recommended quick wins.
7. Optional: request an Atomic Assessment.

Summary table of the templates

Template	Name	When it applies	Output	Release
B1	Company discovery	Proposed to every new workspace	Profile version 1 approved; next workflows proposed	v1
B2	Product deep-dive	One per main product family	Product one-pager; profile product entry updated	v1
B3	Market and product-market fit	After discovery	Market and PMF report	v1 (planning board link is v1.1)
B4	Manufacturing and AI readiness	After discovery	Use-case candidates on the road map	v1
B5	Business plan	When the company needs a plan (bank, customer, programme)	Business plan document	not listed separately; uses canvases (v1.1)
B6	Export readiness	When target markets exist	Export readiness memo	not stated in the SRS

Template	Name	When it applies	Output	Release
B7	AI use-case charter	When defining a use case	Charter document and road map placement	v1 (readiness step is v1.1)
B8	Lite assessment	Self-service pre-audit	Preliminary readiness summary	v1.1

The "Release" column for B5 to B7 is this guide's reading of the SRS (section 11 and the story priorities); the SRS does not give a release per template.

What matters most

- Only B1 to B4 are named in `US-D01` as the AI's first proposals. B5 to B8 are template-driven or user-triggered.
- Notice the repeating shape: question set, optional upload, (AI) analysis, review, output. A single workflow engine with these five step types can run all eight templates.
- Running workflows keep their template version when a template is updated (`US-B05`).

Appendix C: sample question bank

What it is: three samples of the banks the **Interviewer skill** draws from. The AI **picks and rewords** questions according to the gaps and earlier answers. These are sample content, not a fixed form.

B1, Company discovery

- How many people work at the company, and how many of them on the shop floor?
- Which plants or sites do you run, and what does each make?
- Which certifications do you hold, and since when?
- Who are your typical customers: OEMs, tier suppliers, distributors or end users?
- What has changed most in your business in the last three years?

B4, Manufacturing and AI readiness

- Which of these stages exist in your plant: goods receipt, warehouse, kitting, assembly or machining, final inspection, packaging, dispatch?
- At each stage, how is the result recorded: on paper, in a spreadsheet, in a system or by a sensor?

- Which machine produces data that nobody looks at?
- Which decision in the plant takes longest to reach, and who makes it?
- What is counted by hand today that a system or a camera could count?
- When a defect is found, how long is it until the cause is known?

B5, Business plan

- What is the plan for, and who will read it?
- What investment are you planning, and when?
- What do you expect it to change: capacity, quality, lead time or new products?
- How will it be financed, and over how many years?
- Which assumptions worry you most?

What matters most

- Questions are meant to be short and conversational, matching `US-D03` (5 to 10 per set, adaptive).
- The B4 bank is where the **stage-by-stage, how-is-it-recorded** idea starts; it feeds the use-case matching (paper vs spreadsheet vs system vs sensor drives readiness).

Appendix D: document templates in v1

What it is: the **12 document templates** available in the document studio in v1 (the Generate tab, `US-F03`, `US-F08`).

1. Company introduction
2. Business plan
3. Product one-pager
4. Market analysis
5. Product-market fit report
6. Export readiness memo
7. Competitor comparison
8. Programme or subsidy application narrative (the company's description of itself and its plan, in the form the Operator's programmes ask for)
9. AI use-case charter
10. Factory AI road map
11. Customer proposal
12. Meeting preparation brief, for the company or a Coordinator before a meeting

Link to workflows: several workflow outputs in Appendix B produce these documents: B2 gives the product one-pager, B3 the market and product-market fit report, B5 the business plan, B6 the export readiness memo, B7 the AI use-case charter. The road map export (`US-H07`) corresponds to the Factory AI road map.

What matters most

- Templates exist in **both Japanese and English** (`US-F08`).
- Document type names here are what the Generate tab's template dropdown should offer.

Appendix E: traceability from the brief to the stories

What it is: a table linking each of the nine brief items (section 1.2) to the user stories that deliver it. Use it to check nothing in the brief has been left out.

Brief item (section 1.2)	Delivered by
1 Super-admin access and logins for new companies	`US-A01` to `A03`, `A10`; `US-B01`, `B02`, `B08`
2 Website and product details analysed by the AI	`US-C01` to `C03`, `C07`
3 Follow-up questions and follow-up workflows	`US-D01` to `D09`; `US-E07`
4 The company's AI-generated page	`US-C04` to `C06`, `C08`, `C09`
5 Many chats, seeing the workflow, talking to the AI	`US-E01` to `E10`; `US-D05`
6 Knowledge base, prompts to documents, a copilot for business planning	`US-F01` to `F10`; `US-G01` to `G05`
7 Feedback everywhere	`US-I01` to `I07`
8 Use cases from the audit workflow, product discovery and product-market fit, creating your own	`US-H01` to `H10`
9 Japanese first, English equal, across dashboards, chats and generative AI	`US-K01` to `K11`; `US-A09`; `NFR-13`

Stories not mapped to any brief item: `US-A04` to `A08`, `US-B03` to `B07`, `B09`, `B10`, `US-C10`, `US-J01` to `J06`. These are supporting stories (security, admin tools, settings, notifications) that make the brief items work but were not named in the brief. This list is this guide's own reading of the table.

Appendix F: localisation glossary of stages and terms

What it is: the stage names and method terms that need **one agreed Japanese equivalent each before launch**. The equivalents are agreed with the Operator's reviewers (`OQ-10`) and stored in the **localisation resource** and the **workspace glossary**; the SRS itself stays in English for the development team.

Term	Where it is used	Note for the Japanese equivalent
Goods receipt (inward)	Factory map, library, lite assessment	Use the term the member companies' own documents use; "inward" is Atomic Loops' label.
Incoming inspection	Library	Standard quality-management term.
Warehouse	Factory map, library	Plain term.
Kitting	Factory map, library	Often a loan word in Japanese plants; confirm with the Operator.
Assembly; Machining	Factory map	Machining is the variant of the assembly stage shown to job shops.
In-process inspection	Library	Standard term.
Final inspection (pre-delivery inspection)	Factory map, library	The deck's abbreviation PDI is spelled out everywhere.
Packaging	Factory map	Plain term.
Dispatch and logistics	Factory map	Plain term.
Production planning and supply chain	Lane on the factory map	Plain terms.

Term	Where it is used	Note for the Japanese equivalent
Quality; Maintenance	Lanes on the factory map	Plain terms.
Predictive maintenance	Library	Established term.
R&D and product discovery	Lane on the factory map	"Product discovery" needs an agreed rendering; product planning is the usual neighbour.
Product-market fit	Lane, planning canvases	Usually a loan word; confirm.
Sales	Lane on the factory map	Plain term.
Gemba walk	Method explainer	Japanese in origin; use the plant's own everyday term.
Quick win; Strategic bet; Disruptive solution	Method explainer, road map	Three agreed terms, used identically in the interface and in exports.
Autonomy level L0 to L5: Manual, Assisted, Partial, Conditional, High, Full	Method explainer, use-case cards	Keep the L numbers; translate the six names.
Readiness; Entry point (software only, camera based, machine IoT)	Use-case cards, road map	Agreed terms.
Company profile; Workflow; Knowledge base; Road map; Feedback	Navigation	Agreed terms; some may stay loan words.
Stated; Sourced; Inferred	Provenance labels everywhere	Three short agreed terms; the AI uses them in its prose too.

Plain-words reading of a few entries

- **Gemba** is a lean-manufacturing word (from Japanese) for "the real place", meaning the shop floor. A **Gemba walk** is a visit to the floor to observe. (Background explanation, not in the SRS.)

- **PDI** is the deck's short form for pre-delivery inspection, spelled out as "Final inspection (pre-delivery inspection)" everywhere in the product.
- **Machining** is the label shown instead of "Assembly" to job shops (shops that make parts to order rather than assemble products).
- **Loan word** means a foreign word written in Japanese script (katakana). Several terms may simply stay as loan words; the Operator confirms.

What matters most (appendices overall)

- The library seed (A) and the workflow templates (B) are the **content** that makes the product useful on day one. Load them as data (editable in the admin console), not as code.
- Appendix F means there is a **glossary mechanism** in the product: a shared list of terms used by the UI, the AI and exports.
- The L0 to L5 names are fixed in order: Manual, Assisted, Partial, Conditional, High, Full.